Boundaries, Representation, and Local Growth: How Electoral Redistricting Shapes Political Candidacy and Economic Development

Coleson Weir*

October 15, 2026

Job Market Paper
Click here for the latest version

Abstract

This paper examines how electoral redistricting affects political candidacy, representation, and economic development. I utilize India's 2008 Delimitation, which was the world's largest redistricting exercise, as a natural experiment. Redistricting generates constituency-level variation in voter composition and changes in which seats are reserved for candidates from historically disadvantaged social groups. I find that parties nominate more incumbents and fewer previous runners-up in constituencies with more new voters; these constituencies see no altered provision of public goods. Changes in the reservation status of a constituency reduce both incumbent nominations and the probability that an incumbent wins reelection; these constituencies see increased public goods provision and higher nighttime lights growth in the years following redistricting. The findings suggest redistricting boosts local economic development only when it constrains parties' ability to protect incumbents.

Keywords: Electoral redistricting, incumbents, public goods, luminosity growth, India

JEL Classification: P16, D72, O12, O43

*Department of Economics, University of Notre Dame (cweir2@nd.edu). For their excellent research assistance, I thank Adam Battalio, Sagara Ann Johny, Nicole Steiner, Maya Reddy, and Ahmad Wahdat. I thank the Kellogg Institute for International Studies and the Building Inclusive Growth Lab at the University of Notre Dame for financial support. For comments and suggestions, I thank Pascal Frank, Derek Wu, and participants of the Gender and Representation Workshop and Development Economics Workshop at Notre Dame. I extend a special thanks to Lakshmi Iyer, Taryn Dinkelman, Susan Ostermann, and Enrique Seira Bejarano for their feedback and advising on this project.

1 Introduction

Democracies regularly undergo electoral redistricting to equalize population across electoral constituencies to ensure the principle of "one person, one vote." Despite this technical justification to redraw boundaries, redistricting is inherently political. Redrawing electoral boundaries can reshape who runs for office, who wins, and how different communities are represented in the legislature. Understanding how these boundary changes affect political representation is central to evaluating whether redrawing boundaries of electoral districts serves its stated democratic purpose. As elected officials shape local policy and public goods provision, changes in who holds office can carry consequences for local economic development. Evaluating how redistricting alters political representation and downstream economic welfare is necessary to understand the full consequences of how democracies redraw their electoral maps.

New electoral boundaries force candidates and parties to reevaluate their electoral strategies, reshaping the political calculus of a constituency as they compete to win. Parties have long understood this and sought control over the redrawing process, which has historically been a tool to entrench electoral majorities in legislatures (Erikson, 1972). In response, 16 democracies including Canada, India, Mexico, and the United Kingdom have handed the task to an independent commission or agency in order to insulate the boundary-drawing process from partisan actors. As of 2025, seven U.S. states utilize independent commissions to redraw state legislative maps. Even when partisan control is stripped from the redistricting process, the act of redrawing boundaries can still reshape who runs and who is elected. There are also consequences for which places prosper economically, in the wake of these changes. Thus, neutrality in process is no guarantee of neutrality in consequence. As democracies strive for representative governance, understanding how institutional reconfigurations shape both representation and downstream economic development is essential.

In this paper, I analyze how electoral redistricting affects political representation and economic development in India. Following the 2001 census, India redrew electoral boundaries which went into effect in 2008. This was the largest independent redistricting exercise in the world; over 242 million citizens were drawn into new constituencies. Boundaries were redrawn by the Delimitation Commission, which was comprised of non-partisan individuals, whose goal was to equalize population across constituencies. The commission used administrative guidelines to redraw boundaries, so how much any given constituency's boundaries moved is a mechanical byproduct of the pre-existing population distribution. Thus, the extent of boundary changes is plausibly unrelated to the political and economic trajectory of a constituency. The new maps were also not biased towards one political party (Iyer, Weir and

Ostermann, 2026), and did not disadvantage religious minorities (Jensenius, 2013). As redistricting could shape economic development through multiple channels, I develop a conceptual framework outlining these channels to motivate the empirical analysis. Redistricting changes the voter composition of a constituency. Politicians do not then know how a constituency will vote, and voters do not know the politician's type. I call this lack of information uncertainty. In response, political parties may change their electoral strategy in order to maximize their chances of winning a seat. Redistricting also changes who is legally allowed to run for office in India by switching which seats are reserved for candidates from Scheduled Castes and Scheduled Tribes.[1] This provides a new constraint on who political parties may nominate in the election. Altering the candidate pool changes who will ultimately be elected. The preferences of politicians can influence which policy is implemented (Besley and Coate, 1997; Osborne and Slivinski, 1996). Additionally, shifting who the median voter is in a constituency can affect what policy is implemented by the elected representative (Downs, 1957).

I utilize a difference-in-differences identification strategy that leverages the spatial variation in how much a constituency is affected by redistricting. Matching constituencies before and after maps are redrawn is an arduous process. To overcome this obstacle, I overlay GIS maps of state assembly constituencies before and after redistricting with the 2001 population census. Due to some boundary errors in the GIS maps, I manually assigned over 1,200 villages to constituencies. I match constituencies before and after electoral redistricting based on population displacement. This allows me to compute two treatment measures: the fraction of new voters and changes in which seats were reserved for disadvantaged social groups. Including both of these treatment measures separates the uncertainty caused by redistricting and legal restrictions on who can run for office. By comparing changes in constituencies that experienced a high proportion of new voters to those with minimal boundary alterations, I isolate the causal impact of redistricting from confounding state-level trends.

I find that redistricting increases major party incumbent nominations by 14.6% when a constituency has a high proportion of new voters. I show that this is due to political parties changing their electoral strategy. When facing new voters, parties are more likely to nominate incumbent politicians and less likely to nominate previous runners-up. On the other hand when a constituency's reservation status changes, it reduces incumbent nominations by 44.4% due to parties having more limited ability to strategize nominations. These results are consistent with the uncertainty and legal restrictions channels through which redistricting can affect politicians: a high proportion of new voters changes how parties strategize

[1]Scheduled Castes and Scheduled Tribes are historically disadvantaged social groups. At the state level, the seats reserved for these groups only change due to redistricting.

about reelection without barriers on who can run, whereas a change in reservation status can render the incumbent ineligible to hold the seat which limits a political party's strategic ability. While parties do not adjust their nomination of women or religious minority candidates in response to new voters in a constituency, changes in reservation status lead parties to nominate fewer religious minority (Muslim) candidates. These shifts in candidacy carry through to electoral outcomes. Incumbents are marginally more likely to win reelection in constituencies with a high proportion of new voters, and 43.8% less likely to win when their constituency's reservation status changes.[2] The magnitudes of these effects are very similar to those of the candidacy results, suggesting that changes in incumbent candidacy translate directly to changes in an incumbent's probability of winning reelection. Reservation status change reduces the probability a religious minority candidate wins by 47.1%, which is similar in magnitude to the decline in religious minority nominations. The religious minority candidacy effects are concentrated entirely in constituencies moving from unreserved to reserved for disadvantaged social groups. Constituencies that became reserved also see increases in women's candidacy. This suggests there are tradeoffs in affirmative action policies. Affirmative action can have positive spillovers to some underrepresented minorities, but also crowd out other minority groups.

As politicians are able to affect public policy, this paper goes on to examine how electoral redistricting changes economic performance and development. Previous literature has examined the impact of incumbent politicians as well as women politicians, and has found mixed results. Entrenched incumbents typically worsen economic development and voter well-being (Acemoglu and Robinson, 2006; Ashworth, 2006; George, 2024); however, political turnover may improve or worsen public service delivery (Akhtari, Moreira and Trucco, 2022; Bazzi et al., 2025; Marx, Pons and Rollet, 2025; Toral, 2024). Therefore, the impact of electing new politicians on economic development is ambiguous. Another way different politicians alter policy is through their identity. Women are more responsive to their constituents and bring back more money to their constituency than male politicians (Anzia and Berry, 2011; Bauer and Cargile, 2023; Thomsen and Sanders, 2020), but electing women may or may not change policy (Chattopadhyay and Duflo, 2004; Clayton, 2021).

Using the same identification strategy, I find that redistricting generates gains in local economic activity but only when there is a change in political representation. When a constituency has a high proportion of new voters, the growth in primary school provision slows. There is no effect of new voters on other public goods provision or economic activity. Changes in reservation status increase middle school provision by 4.4% and increase primary health center provision by 11.7%. These same constituencies have faster nighttime lights

[2]The representation outcomes are not conditional on being nominated.

growth of 1.23 percentage points. Redistricting has one third of the effect of programs directly targeting electrification in India (Burlig and Preonas, 2024). A back of the envelope calculation suggests the 1.23 percentage point increase in luminosity growth corresponds to a 0.18 percentage point increase in GDP growth. This is an estimated 2.26 billion dollars in gained economic activity per year over a politician's five-year term. The improvements in public goods provision and nighttime light growth are seen in both constituencies that became unreserved and constituencies that became reserved. This suggests that changes in political representation can improve economic development for constituents.

This paper contributes to the literature on political selection and the barriers to representation by showing that institutions can alter candidate entry, even when the institution is designed to be independent. Standard economic models of political entry suggest that candidates decide to run based on their expected probability of winning, the cost of campaigning, the returns to holding office, and electoral rules (Arora, 2022; Beath et al., 2016; Lawless and Pearson, 2008). While prior work has focused on individual-level constraints such as risk aversion or supply-side barriers (Gulzar, 2021; Lawless, 2015; Norris and Inglehart, 2001), less attention has been paid to how the physical boundaries of the electorate alter these parameters. Redistricting shapes the candidate selection environment by altering the spatial distribution of voters and potentially disrupting the political capital built by incumbents. By analyzing this institutional shock, I show that independent boundary changes can further entrench political power unless redistricting is paired with changes in who is eligible to run for office. Affirmative action is one way to counteract entrenching political power. Reserved seats for historically disadvantaged groups also have positive spillovers for women's candidacy and negative spillovers for other religious minority candidates.

This paper also adds to the literature examining the economic and political consequences of malapportionment and electoral redistricting by showing that even neutral and independent redistricting processes can still affect political candidacy and development outcomes. When constituencies have unequal populations, this leads to unequal representation, development, and economic policy (Ansolabehere, Gerber and Snyder, 2002; Ardanaz and Scartascini, 2013; Bhavnani, 2018, 2021). Redistricting aims to equalize population to counteract malapportionment. While prior research has focused on the theoretical properties of optimal gerrymandering (Friedman and Holden, 2008; Gul and Pesendorfer, 2010) or the spatial compactness of districts (Fryer and Holden, 2011; McKinnie and Szalda-Petree, 2024), this paper provides the first causal evidence on how redistricting reshapes candidate selection and subsequent local economic development, in a setting where redistricting was conducted independently of political actors. This isolates the effects from partisan gerrymandering. Most of the literature focuses on gerrymandering in the United States. The papers that uti-

lize microdata in the U.S. have primarily examined on the effects of redistricting on voters, voter turnout, and party vote shares (Fraga, 2016; Hayes and McKee, 2009; Weir, 2026*a*), as well as women's representation (Weir, 2026*b*). Only a few papers examine redistricting in a developing country. In India, studies have examined the process itself (Iyer, Weir and Ostermann, 2026; Jensenius, 2013) and its effect on infant mortality (Kjelsrud, Moene and Vandewalle, 2024).[3]

The rest of the paper is structured as follows: Section 2 covers the institutional background and Section 3 discusses the conceptual framework. Section 4 explains the data used in the analysis, and Section 5 describes the empirical strategy. Section 6 contains the results for political candidacy and representation, while Section 7 explores the economic effects of redistricting. Section 8 concludes.

2 Institutional Background

2.1 India's Electoral System

India is a multiparty parliamentary democracy at both the national and state levels. Six states have bicameral legislatures while the other 22 states have unicameral legislative bodies.[4] Each state holds first-past-the-post elections approximately every 5 years. The elections are competitive with about eight candidates running in each race, on average. Candidates are not prohibited from running in more than one constituency; a small number of candidates decide to run for multiple seats at the same time. State legislative elections are conducted by the Election Commission of India which is comprised of career bureaucrats who may not participate in partisan activities, including joining political parties. Elections are held in phases to allow for proper election security measures.

Across all levels of government, India has reserved legislative seats for historically underrepresented groups. Articles 330, 332, and 334 of the Constitution of India create reserved seats in the Lok Sabha and state legislative assemblies for Scheduled Castes (SC) and Scheduled Tribes (ST).[5] Scheduled Castes are groups that have historically held the lowest position in the Hindu caste system, whereas Scheduled Tribes are communities that historically existed outside of the caste system altogether. The proportion of seats reserved should be approximately the same as the proportion of the SC and ST population in each state

[3]Research has also examined how administrative unit splits impact development, and found mixed results (Cassidy and Velayudhan, 2026; Lipscomb and Mobarak, 2017; Narasimhan and Weaver, 2024).

[4]Andhra Pradesh, Bihar, Karnataka, Maharashtra, Telangana, and Uttar Pradesh have bicameral state legislatures.

[5]The Lok Sabha is India's lower chamber of Parliament.

and union territory. There are not currently reserved seats for women in state legislatures nor the Lok Sabha; however, the 106th Amendment reserves one-third of seats in the Lok Sabha and state legislative assemblies for women, which will go into effect following the next delimitation.[6]

2.2 History of Delimitation in India

Every few years, 42 democracies redraw the boundaries of single-member districts.[7] Less than half of these countries have independent commissions that redraw boundaries. In India, the Delimitation Commission is responsible for conducting electoral redistricting, or delimitation as it is called in India, and was outlined in Article 82 of the Constitution of India. The Delimitation Commission of India is designed to operate as an independent, nonpartisan body to insulate the redistricting process from political interference. Under the Delimitation Act of 2002, the commission consisted of three core members with voting power: a retired judge of the Supreme Court of India, who served as the chairperson; the Chief Election Commissioner of India; and the Election Commissioner of the respective state or union territory. To ensure local context is considered, the commission enlisted ten associate members from each state, consisting of five members of the Lok Sabha and five members of the state legislative assembly.[8] The associate members did not possess voting rights, nor did they have the power to veto or sign the final boundary determinations. The final authority to approve and publish the electoral maps rested solely with the three members of the commission, which significantly limited the scope for partisan gerrymandering or strategic boundary manipulation by political incumbents.

Article 170 of the Constitution of India lays out the principle of "one person, one vote" and ensures the proportion of the population contained within each constituency should be approximately the same within each state. The process of electoral redistricting occurred following each decennial census until states became concerned that redistricting unfairly redistributed political power across states due to unequal progress in family planning policies. The concerns resulted in the 42nd Amendment, which said the 1976 electoral boundaries would be used until after the first census following the year 2000. The population census was conducted in 2001, and the process of redrawing electoral boundaries began with the Delimitation Act of 2002. States were still concerned about reallocating seats across states due to differing population growth in northern and southern states. As a result, the number

[6]The next delimitation is supposed to occur following the 2027 census.

[7]All but nine of the 42 countries that redraw boundaries are historically linked to the United States, United Kingdom, or one of their former colonies (DeSilver, 2025).

[8]If a state had fewer than five members of the Lok Sabha, all members of the Lok Sabha from the state served as advisory members of the delimitation commission.

of seats in both the Lok Sabha and state legislative assemblies were kept the same, and seats were only reallocated within states.

In redrawing constituency boundaries, the Delimitation Commission followed specific guidelines to ensure geographical compactness and administrative continuity. Constituencies must be contiguous geographical areas, and the commission was instructed to utilize natural barriers, public roads, and existing communication links as boundaries. Detailed empirical analysis of whether there was political influence in the redistricting process in India by Iyer, Weir and Ostermann (2026) shows politicians were unable to influence redistricting in their favor.

The distribution of reserved seats for Scheduled Castes and Scheduled Tribes followed constitutional regulations that minimized administrative discretion. According to the Delimitation Act of 2002, constituencies with the highest concentration of Scheduled Tribe populations were reserved for Scheduled Tribes. For Scheduled Castes, the commission was required to identify constituencies with highly concentrated Scheduled Caste populations and distribute the reserved seats geographically across different regions of the state to ensure broad representation. Because these assignments were determined mathematically based on the population shares recorded in the 2001 census, the shifting of a constituency into or out of reserved status is a mechanical consequence of demographic changes rather than political negotiation. This assignment provides a source of variation in candidate eligibility that is independent of local politics.

Using the 2001 census data, the delimitation commission created a draft of the new boundaries for each state. Draft reports were published for each state between 2004 and 2007. Comments on the proposed boundaries written by advisory members of the commission were published alongside the draft reports. The commission gathered feedback on the proposed boundaries through public comments and public meetings. After taking the feedback into account, the Delimitation Commission published final boundary reports for each state. The new maps drawn by the Delimitation Commission were implemented for state and national elections beginning in 2008. These boundaries are still in effect and will be until after the first election following the 2027 census.

The goal to equalize population across constituencies was generally achieved. Redistricting shifted the population distribution of constituencies, and reassigned reserved seats across constituencies. Constituencies have a higher population, on average, and the standard deviation shrank by 22% (Appendix Figure A.1). Overall, the goal of redistricting was achieved by the Delimitation Commission.

3 Conceptual Framework

Electoral redistricting can affect policy outcomes through two main channels: by changing the composition of voters within a constituency and by changing who is elected to represent a constituency. I discuss each channel in turn. Appendix Figure A.2 presents a diagram of these two channels and the mechanisms linking them to policy and development outcomes.

Redistricting redraws constituency boundaries, altering the underlying population of voters that a politician must appeal to. A new composition of voters can shift policy outcomes by changing the median voter in a constituency, since the median voter's policy preference is the one implemented in equilibrium (Downs, 1957). If redistricting moves a constituency's median voter either by adding or removing villages with different preferences, the policy that is implemented after the election shifts accordingly.

This channel operates not only through the redrawing of boundaries but through its effect on voter turnout. Voters who are placed into a new constituency are less likely to vote (Hayes and McKee, 2009), meaning their preferences may not be taken into account even though they are represented. If redistricting dampens voting turnout, the median voter of those who vote may not coincide with the median voter among the full constituency population. A politician optimizing their election chance responds to this effective median voter. Thus, redistricting can change policy through changing voter turnout and thus the effective median voter of a constituency. A limitation of this channel is that voter preferences are not directly observable in the data. As a result, I cannot directly test whether redistricting shifts the median voter or the effective median voter in a constituency. However, I do test whether redistricting changes voter turnout.

The second channel operates through redistricting's effect on political candidacy, and in turn on who wins political office. Redistricting creates a constituency with a new electorate. Politicians do not know how a newly drawn constituency will vote. On the other hand, new voters do not know the incumbent politicians or their political record. I refer to this information problem as uncertainty. *Ex ante*, it is not clear how this uncertainty would affect candidacy. On one hand, parties may respond by favoring their incumbents (Meserve, Pemstein and Bernhard, 2020). Incumbents have already proven they can win elections, and parties tend to nominate candidates likely to win (Magesan, Szabó and Ujhelyi, 2025). Therefore, redistricting could increase the nomination of incumbents and previously elected politicians, and correspondingly reduce the nomination of other previous candidates, women, and religious minority candidates. On the other hand, parties may not nominate incumbents for reelection if the incumbent is not a good fit for the voters in the new constituency. If fewer incumbents are nominated, this may open up opportunities for other candidates. I am

going to examine whether parties nominate incumbents, and what other candidates may run instead. In addition to incumbents, I focus on two types of candidates that are historically underrepresented, women and religious minority candidates, and whether this geographic reorganization opens doors to these candidates.

This same uncertainty can affect individual candidates' decisions to run, independent of party nomination strategy. Incumbents may decide to run in multiple constituencies to hedge their bets and increase their odds of retaining a seat in the legislature. Redistricting may also make reelection more attractive. As newly added voters are less familiar with the incumbent and their record, they may offset the anti-incumbency bias that would otherwise prevail in the absence of redistricting (Uppal, 2009). Uncertainty affecting candidacy decisions may differentially impact women. Women already face higher barriers and costs to running for office (Lazarus and Steigerwalt, 2018; Shaukat, 2019). More generally, running in a constituency whose voting behavior is unknown is inherently riskier, and women tend to be more risk averse; the uncertainty could then reduce women's candidacy directly (Niederle and Vesterlund, 2007; Preece, Stoddard and Fisher, 2016). To test whether individual candidate's decisions are affected by redistricting, I examine how the candidacy decisions of independent candidates changes once new maps are implemented.

A shift in who runs may or may not translate into changes in representation. Incumbents may be less willing to run for a seat in a new constituency where the electorate's voting behavior is unpredictable. If the incumbents who withdraw from newly drawn constituencies are not a systematically stronger or weaker subset of incumbents overall, then the decline in incumbent candidacy should map closely onto the decline in the probability that an incumbent wins the seat, with representation shifting primarily through reduced entry rather than through a change in the competitiveness of the incumbents who do run. This logic may not extend to women and religious minority candidates. Parties may nominate them more often in a new constituency without committing the same support they give incumbents. If so, a rise in their candidacy would not bring the same rise in their odds of winning for women and religious minority candidates.

In the case of India, redistricting also directly changes who is eligible to run by changing a constituency's reservation status for Scheduled Castes and Scheduled Tribes.[9] When a constituency moves from unreserved to reserved, the sitting incumbent can no longer run unless they belong to a Scheduled Caste or Scheduled Tribe. On the other hand when a constituency moves from reserved to unreserved, the incumbent typically does not seek reelection (Bhavnani, 2017). A reservation status change could thus alter the number of incumbents

[9]A constituency's reservations status can only change due to the delimitation process for state assembly seats in India.

running post-redistricting. Because non-political reservations for SC and ST individuals have generated gendered spillovers elsewhere (Bertrand, Hanna and Mullainathan, 2010; Cassan, 2019), reservation status changes from redistricting may similarly carry unintended benefits or consequences for women's candidacy.[10]

Changes in political candidacy and electoral outcomes may shape economic development through three channels. First, changing who runs in a constituency can alter the political competition in the election. Increases in political competition can lead to economic growth through the adoption of more efficient policies (Ashworth et al., 2014; Besley, Persson and Sturm, 2010; Padovano and Ricciuti, 2009). The second channel is through the election of women to office which results in higher levels of economic growth (Baskaran et al., 2024), investment in education (Clots-Figueras, 2012), and public goods provision (Chattopadhyay and Duflo, 2004). These gains, however, are conditional on women's ability to exercise effective authority once elected. When inexperience or informal proxy control by male relatives leaves women's seats susceptible to capture, the anticipated benefits of female representation fail to materialize (Afridi, Iversen and Sharan, 2017). The last channel is through incumbents. Whether incumbents are beneficial or detrimental to development is ambiguous. Entrenched incumbents are associated with weaker electoral discipline and correspondingly lower growth and public goods provision (George, 2024), suggesting that turnover among such incumbents should be development-enhancing. Yet this turnover can induce bureaucratic turnover, which is disruptive to administrative processes, and degrade service delivery (Akhtari, Moreira and Trucco, 2022; Toral, 2024). Where a change in a constituency's reservation status results in political turnover, service delivery may therefore deteriorate even as electoral accountability improves.

4 Data

4.1 Measuring Electoral Redistricting

To measure the extent of electoral redistricting, I utilize data from Iyer, Weir and Ostermann (2026). At the village level, India's 2001 Population Census data is overlaid with GIS maps of state assembly constituency boundaries before and after the 2008 delimitation. To resolve spatial discrepancies arising from boundary misalignment in the raw GIS shape files, particularly along international borders and coastal regions, manual matching for approximately 1,200 villages was required. Each disputed village was matched using its census

[10]Studies have examined the effects of reserved seats in the local setting, with mixed results (Cassan and Vandewalle, 2021; Karekurve-Ramachandra and Lee, 2020).

land area or centroid coordinates. When a village polygon intersected multiple constituency boundaries due to digitization errors, the village was assigned to the constituency containing its administrative center or the largest share of its land. 23 states in India underwent redistricting in 2008 and are thus included in the analysis.[11] Pre-redistricting constituencies are matched to post-redistricting constituencies based on whichever post-redistricting constituency contains the largest proportion of the pre-redistricting constituency's population.[12] For example, in Andhra Pradesh, Madakasira constituency stayed the same, Kalyandurg constituency shrunk, and Kadiri constituency lost some land in the northwest but gained land in the south (Appendix Figure A.3). To illustrate this, constituency maps before and after redistricting for Anantapur district in Andhra Pradesh are presented in Appendix Figure A.3.

This methodology allows me to compute two main measures of redistricting: the fraction of new voters and reservation status change. The fraction of new voters in a constituency is computed as $1 - \frac{\text{Old Constituency Population in New Constituency}}{\text{New Constituency Population}}$. This is computed for where new constituency that contains the largest proportion of the old constituency's population. Constituencies have approximately a third of constituents in the constituency that were not before redistricting, on average (Table A.2).[13] The median proportion of new voters in a constituency is 30.7%. 12.5% of constituencies have less than 5% new voters. Appendix Figure A.4 shows the distribution of the fraction of new voters. Redistricting resulted in substantial boundary changes, with many voters being displaced from their prior constituency.

Redistricting changes which seats are reserved for SC and ST candidates. I compute whether a constituency had a change in its reservation status. This includes both a constituency going from unreserved to reserved for Scheduled Caste or Scheduled Tribe candidates from reserved to unreserved. Moving from unreserved to reserved would legally bar a politician from running in that constituency if they are not a member of a Scheduled Caste or Scheduled Tribe. Alternatively, if a candidate goes from reserved to unreserved, a SC/ST politician can still run in the constituency, but they are unlikely to win. Approximately 20%

[11] A few cities had the boundaries and number of wards change substantially. Thus, constituencies for these cities were unable to be matched and thus dropped from the analysis. This accounts for 6% of the total sample of the 23 states. For various reasons such as safety concerns due to ongoing conflict, Arunachal Pradesh, Assam, Jammu & Kashmir, Jharkhand, Manipur, and Nagaland did not redraw constituency boundaries in 2008. New boundaries for Jharkhand were drawn, but not implemented. Assam did not redraw constituency boundaries in 2008, but did in 2023.

[12] Other research examining boundaries or boundary changes typically look at whether boundaries split certain groups of people (Bluhm, Hodler and Schaudt, 2024; Michalopoulos and Papaioannou, 2016) or measure the artificialness of a boundary (Alesina, Easterly and Matuszeski, 2011).

[13] There are 66 constituencies with more than 80% of their constituency being new voters. Many of these constituencies became much larger due to redistricting. The high fraction of new voters could also be high due to some remaining errors after manual corrections due to imprecise borders of the GIS maps.

of constituencies had their reservation status change due to electoral redistricting (Table A.2).

Another measure of redistricting is the demographic change of a constituency. To measure this, I construct a demographic change index for each constituency. The index includes the population shares of women, scheduled castes, scheduled tribes, literate individuals, and rural residents. The demographic change index is defined as the square root of one fifth of the sum of the squared differences across these five dimensions, which is normalized strictly between zero and one. This metric captures the multidimensional demographic distance between the old and new boundaries, where a value of zero indicates identical demographic compositions and a value of one represents a complete demographic inversion. On average, constituencies in India have 3.9% demographic change, and it ranges from no change at all to 47% change. The correlation between the demographic change and the fraction of new voters is 0.355. I use the fraction of new voters and reservation status change as the primary measures of redistricting because the demographic change measure is limited by what is recorded in the population census.[14]

4.2 State Legislative Election Data

I utilize election data for India's state assemblies from the Trivedi Centre for Political Data. The sample includes state assembly elections from 1978 through 2017.[15] The data contains information on political party, sex, and the number of votes received for each candidate.[16] I merge these data with data identifying which candidates are Muslim based on their names. Muslim candidates are classified as religious minority candidates for the remainder of the paper. The religion data is available for all elections before redistricting and the election immediately following redistricting. Overall, there are 202,646 unique candidates who ran for election across the 3,235 constituencies. Of the 23,487 incumbents, 1,309 are women, and 22,178 are men. The elections are competitive even when incumbents are present. In the election immediately before redistricting, 8.4 candidates run for office in each constituency. Less than 2% of candidates decide to run for multiple seats in the same election year. Major parties nominate 0.61 incumbents on average before redistricting, which drops to 0.48 after redistricting.[17] Incumbents win 31% of seats prior to redistricting, which drops to 25% after new maps are implemented. On the other hand, women win only 7% of legislative assembly

[14] Age, education, income, and granular occupational categories are not in the population census data.

[15] For specific election years by state, see Appendix Table A.1.

[16] Each candidate's age and caste is not available for all individuals. Caste is an optional question on candidate affidavits, and few candidates report this information.

[17] Major political parties are defined as a party that held at least 5% of legislative seats in any state between 1978 and 2008 following Bhalotra, Clots-Figueras and Iyer (2018)

seats, and religious minority candidates only win 5% of seats before redistricting (Table 1). Women and religious minority candidates' probability of winning marginally increases after redistricting. The data contains a unique candidate identifier to track candidates over time. Utilizing the candidate identifier, I determine if candidates seek office in the next election and whether they win. About 63% of incumbents run for reelection prior to redistricting, while almost half of these incumbents win the next election.

Table 1: Summary Statistics

	Pre-Redistricting Mean (1)	Post-Redistricting Mean (2)
Panel A: Political Candidacy		
# Incumbents Nominated by a Major Party	0.657	0.633
# Women Nominated by a Major Party	0.329	0.405
# Minority Candidates Nominated by a Major Party	0.338	0.377
# Independent Men	3.014	3.776
# Independent Women	0.180	0.234
Panel B: Voter Behavior and Representation		
Voter Turnout	0.710	0.749
Any Incumbent Won	0.336	0.326
Woman Won	0.070	0.085
Minority Candidate Won	0.051	0.054
Independent Won	0.038	0.027
Panel C: Economic Development		
Primary Schools	87.251	88.632
Middle Schools	39.112	51.940
Primary Health Centers	7.702	8.311
Primary Health Subcenters	22.317	30.615
Paved Roads	68.635	73.523
Annual Nighttime Light Growth	7.484	4.746

Notes: This table presents summary statistics of all of the main outcomes. Pre-redistricting data is for the election immediately before redistricting. Post-redistricting data is for the election immediately after redistricting. The public goods in Panel C are computed as the percent of villages in a constituency with a specific public good.

4.3 Economic Data

I utilize the Socioeconomic High-resolution Rural-Urban Geographic Platform for India (SHRUG) for data on night lights, public goods, employment, and pollution (Asher et al., 2021). Using unique village-level identifiers from the SHRUG, I spatially match the villages

to assembly constituencies before and after the 2008 redistricting.[18] Pre-redistricting constituencies are matched to post-redistricting constituencies based on the mapping described above.

4.3.1 Public Goods

The Population Censuses from 1991, 2001, and 2011 contain the number of specific public goods in each village. Using this data, I compute the percent of villages within a constituency that have the following public goods: education facilities, health facilities, and paved roads. For education and health, I examine primary schools, middle schools, primary health centers, and primary health subcenters. Public goods provision in India has been increasing since the 1961 Census. There have been many public policies aimed at ensuring everyone has access to public goods such as schools and health centers (Iyer and Weir, 2025). In 1991, 84% of constituencies had a primary school, which increased to 89% in 2011. Similarly, for primary health subcenters, 26% of constituencies had one in 1991, and 31% had a primary health subcenter in 2011. There was a larger increase in the provision of paved roads over the same two decades. The provision of paved roads increased by 11 percentage points from 1991 to 2011.

4.3.2 Nighttime Lights

I use nighttime light data as a proxy for economic growth. A benefit of using night light data as a proxy is that it allows for estimates at a disaggregated level, where GDP is not commonly computed (Henderson, Storeygard and Weil, 2012). Baskaran et al. (2024) find that a 1 percentage point increase in night lights is associated with a 0.15 percentage point increase in GDP in India. Night lights themselves are associated with electricity usage in an economy (Chand et al., 2009), and electrification improves economic development outcomes (Dinkelman, 2011; Lipscomb, Mobarak and Barham, 2013). In India, politicians can control the distribution of electricity (Baskaran, Min and Uppal, 2015; Burgess et al., 2020; Mahadevan, 2024). Thus, night lights are able to be influenced by politicians and a good proxy for economic growth and development.

[18]This mapping resulted in fewer constituencies than the election data contains due to villages in entire constituencies not being included in the SHRUG data, including the entirety of the state of Puducherry. The aggregated SHRUG data has 2,760 constituencies across the 22 states in my sample (not including Puducherry). Some states in the SHRUG data have fewer constituencies than in the election data. For example in Andhra Pradesh, there are 279 constituencies in SHRUG even though there are 286 in the election data. Other states have more constituencies than the election data due to dropping cities, in my data,that could not have their constituencies mapped. I therefore matched 2,436 constituencies to the SHRUG data for public goods. I conduct a robustness check of the candidacy and representation results using the same sample as the public goods and nighttime light data.

The SHRUG contains nighttime light data from the U.S. Air Force Defense Meteorological Satellite Program's Operational Linescan System (DMSP) (Asher et al., 2021; Henderson, Storeygard and Weil, 2011). The data is available from 1994 through 2013. Each pixel is assigned a value from 0 to 63 depending on the intensity of the light in that pixel. The National Oceanic and Atmospheric Agency adjusted the DMSP data by removing any low quality data. Data could be removed due to environmental reasons such as clouds, auroral lights, and active fires in a given area.[19] This refined data is aggregated by year. I aggregate the village night light data to the assembly constituency level to create yearly estimates of night lights in each electoral constituency. I scale the total sum of night lights in a constituency by the area of the constituency. The data is aggregated to the five-year election terms.

5 Estimating the Impact of Electoral Redistricting

To estimate the effect of electoral redistricting, I use a difference-in-difference identification strategy that exploits the fraction of new voters in a constituency as well as the change in the reservation status of a constituency. I estimate the difference-in-difference model with the following regression equation:

$$y_{ct} = \beta_1 Post_t \times NewVoters_c + \beta_2 Post_t \times ResChange_c + \delta_c + \delta_t + \varepsilon_{ct} \tag{1}$$

where y_{cst} is the outcome of interest in electoral constituency c in election year t. I stack the data such that the election years are relative to the election immediately preceding electoral redistricting. For example, the elections in Karnataka and Andhra Pradesh that immediately preceded redistricting occurred in 2004 and 2005, respectively. These election years are both coded as 0. $Post_t$ is an indicator variable equal to one if the election occurs after redistricting. $NewVoters_c$ is an indicator equal to one if a constituency has the fraction of new voters in the constituency at or above the median across the entire country. $ResChange_c$ is an indicator equal to one if the reservation status of a constituency changed due to redistricting. Including both $ResChange_c$ and $NewVoters_c$ in the analysis separates the legal restrictions on who can run for office from the uncertainty caused by redistricting.[20] Constituency and time fixed effects are denoted δ_c and δ_t. Constituency fixed effects capture both observed and

[19] Low sensitivity and saturation could all introduce measurement error in the night light data. Min et al. (2013) and Tuttle et al. (2014) show that satellite sensors can pick up very low levels of light output, suggesting low sensitivity is not an issue. Saturation could result in the highest light output being censored, as there is a cap on how high light can be measured.

[20] This identification strategy assumes the effects are linearly additive and not multiplicative. The correlation between $ResChange_c$ and $NewVoters_c$ is 0.07, so any multiplicative effect would be incredibly small, if one exists.

unobserved time-invariant constituency characteristics. Relative election year fixed effects control for time trends common to all electoral constituencies. The main analysis uses one election before and one election after redistricting. However, I show robustness to utilizing all election years. Standard errors are clustered at the post-redistricting electoral constituency level to account for a single post-redistricting constituencies being matched to multiple pre-redistricting constituencies.

This empirical strategy avoids issues presented by the difference-in-differences and econometrics literature by not including time-varying covariates (Caetano et al., 2024), not utilizing continuous treatment (Callaway, Goodman-Bacon and Sant'Anna, 2025), and not exploiting variation that affects groups at different times (Goodman-Bacon, 2021).[21] The Delimitation Commission's maps went into effect in 2008 and would be first utilized in the first election after 2008. Elections occur every five years and the dates are predetermined. Thus, all of the maps go into effect in the same year even though the elections can occur at different times. Thus, there is no staggered rollout of the new maps, and the two-way fixed effects estimator is not biased.

For the overall identification strategy to yield a causal estimate, the change in the outcome variable for constituencies with a low proportion of new voters (no reservation status change) would be the same as the change experienced by constituencies with a high proportion of new voters (reservation status change) in the absence of electoral redistricting. To provide evidence for this parallel trends assumption, I will estimate an event study model which allows the effect of electoral redistricting to vary over time. By interacting the fraction of new voters in a constituency as well as the reservation status change indicator with election year fixed effects, I test for preexisting trends and estimate the effect of electoral redistricting on the outcome of interest over time. The event study model is estimated using the following equation:

$$\begin{aligned} y_{ct} = & \sum_{\tau=-6, \tau\neq 0}^{2} \alpha_\tau \mathbf{1}\{ElectionYear = \tau\} \times NewVoters_c \\ & + \sum_{\iota=-6, \iota\neq 0}^{2} \gamma_\tau \mathbf{1}\{ElectionYear = \iota\} \times ResChange_c + \delta_c + \delta_t + \nu_{ct} \end{aligned} \tag{2}$$

where τ and ι represent the leads or lags of interest. The variables $NewVoters_c$, $ResChange_c$, δ_c, and δ_t are as described above. The coefficients α_{-6} through α_{-1} test whether outcomes

[21] I utilize a stacked difference-in-difference identification strategy. Although the treatment does not occur at different times, Baker, Larcker and Wang (2022) show that the stacked difference-in-difference eliminates bias caused by staggered rollout and gives unbiased estimates. The difference between a staggered difference-in-difference and estimators suggested in Callaway and Sant'Anna (2021) and Sun and Abraham (2021) is the weighting of observations (Baker, Larcker and Wang, 2022).

followed common trends in constituencies with high and low proportions of new voters prior to electoral boundaries being redrawn. The coefficients of interest for the fraction of new voters are α_1 and α_2, which measure the effect of a high proportion of new voters on the outcome of interest in the two elections following redistricting. The coefficient for election year 0 is omitted so the other coefficients are all relative to the election immediately preceding electoral redistricting. The coefficients γ_{-6} through γ_{-1} test whether outcomes followed common trends in constituencies that experienced a reservation status change and those that did not prior to electoral boundaries being redrawn. The coefficients of interest for the analysis of reservation changes are γ_1 and γ_2 which measure the effect of a reservation status change on the outcome of interest in the two elections following redistricting. Standard errors are also clustered at the electoral constituency level.

6 Does Redistricting Alter Political Candidacy and Representation?

6.1 Political Candidacy

Changing the voting population of a constituency shapes who runs for office. When there is a high proportion of new voters in a constituency, the number of incumbents nominated by a major party increases by 0.096 (Table 2). This is a 14.6% increase in the number of incumbents running for office, which translates to approximately one more incumbent running in every 10 constituencies. This boost to incumbent nominations is slightly smaller than the incumbency advantage in rerunning in elections post-1991 (Uppal, 2009). Uppal (2009) finds a candidates who barely won an election are approximately 18% more likely to run in the next election than candidates who barely lost. Facing a new set of voters in an election boosted incumbent politician's advantage even more. Table 2 shows that having new voters in a constituency due to redistricting increases the number of incumbent politicians running in the election.

The uncertainty due to new voters did not change the number of women or religious religious minority candidates nominated by major political parties. It also did not change the number of independent men or women running in the election (Table 2). The estimates for each of these are negative, but statistically insignificant. This suggests that while incumbents are benefited by new voters in a constituency, other candidates are not.

Reserved seats legally bar certain people from running for office in a constituency. The number of major party nominated incumbents declines by 0.292 when a constituency's reservation status changes, which is a 44.4% decline (Table 2). In one out of every three to four

constituencies, incumbents are no longer nominated by major parties as a result of a change in reservation status. The effect is roughly three times as large in constituencies that became reserved because incumbents were unlikely legally eligible to run in that constituency (Appendix Table A.3). This reflects the legal difference between the type of change in the reservation status of a constituency.

Women and religious religious minority candidates are treated differently by political parties when parties are faced with legal restrictions on who is able to run for office. Major parties increase the number of women nominated by 0.087 women, on average, due to a reservation status change, although this effect is not significant at conventional levels (Table 2). This increases the number of women running for a major party by 26.4%. On the other hand, religious religious minority candidates are nominated less by major political parties due to changes in which constituencies are reserved for SC/ST candidates. Religious religious minority candidacy decreases by 55.3% when a constituency's reservation status changes (Table 2). The changes in women's and religious religious minority candidate's candidacy are only observed when a constituency moves from unreserved to reserved as a result of electoral redistricting (Appendix Table A.3). In constituencies that become reserved, the number of women nominated increases by 54.4%, while the number of candidates from religious minorities decreases by almost 100% (Appendix Table A.3). The decrease in Muslim candidacy is likely due to Muslims not being included in the caste system and not included in SC/ST reservations. The different effects of affirmative action for certain groups shows that affirmative action can have positive spillovers to as well as crowding out of other underrepresented groups.

The new voters in a constituency and legal restrictions on candidacy changed who ran for public office. Incumbents are nominated more by major political parties due to new voters and less due to changes in the reservation status of a constituency. Women and religious religious minority candidates are only affected by legal restrictions on who is able to run, with parties nominating more women and fewer religious minority candidates when a constituency becomes reserved. These results hold for the total number of incumbents, women, and religious minority candidates as well (Appendix Table A.4).

6.1.1 Why Do Incumbents Benefit From New Voters?

Why do incumbent run more when faced with uncertainty surrounding how a constituency will vote? The number of incumbents running increases by 14.6% (Table 2). Is this due to incumbents running in multiple constituencies to hedge their bets of winning or parties changing their strategy of who to nominate? Table 3 shows that incumbents are only more likely to run in one seat, but not multiple seats. Incumbents are therefore not hedging their

Table 2: Effect of Electoral Redistricting on Political Candidacy

	Major Party Nominees				
	Number of Incumbents (1)	Number of Women (2)	Number of Muslim Candidates (3)	Number of Independent Men (4)	Number of Independent Women (5)
$Post \times NewVoters$	0.096***	-0.019	-0.035	-0.201	-0.055
	(0.037)	(0.043)	(0.049)	(0.177)	(0.034)
$Post \times ResChange$	-0.292***	0.087	-0.187***	-0.077	0.059
	(0.043)	(0.057)	(0.058)	(0.222)	(0.042)
Observations	6,470	6,470	6,470	6,470	6,470
R-squared	0.559	0.577	0.719	0.681	0.534
Avg Dep Variable	0.657	0.329	0.338	3.014	0.180

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

bets to win across multiple places; they are just more likely to run for reelection. If individual decision making was the main driver of why more incumbents are running, one would expect the number of independent candidates running to rise, but this is not the case (Table 2). However, parties reduce the nominations of previous runners up by 21.4% and other previous candidates by 25% (Table 3). This shows that political parties are changing their nomination strategy in response to redistricting. This change in response is only seen in the first election with new maps. Incumbents are no longer more likely to be nominated due to new voters in the second election following redistricting (Figure 1). The effect on runners-up goes back towards zero in the second election, and is null for other previous candidates in that election (Appendix Figure A.5).

Similarly when parties are faced with a reservation status change, parties nominate fewer incumbents as well as previous runners-up and other candidates (Table 3). This is likely due to these candidates not belonging to a Scheduled Caste or Scheduled Tribe. Runners-up and other previous candidates are more effected than incumbent politicians by reservation status changes, as the effect is larger as a fraction of the mean for these types of candidates. When a constituency becomes reserved, the pool of potential candidates shrinks because

candidates must be SC or ST. As a result, parties nominate more women (Appendix Table A.3). Overall, the results show that incumbents are not purely trying to keep their seats. Political parties are changing who they nominate in order to have the best chance of winning an election.

Table 3: Effect of Electoral Redistricting on Incumbents and Party Strategy

	Number of Incumbents		Major Party Nominees	
	Running in One Seat (1)	Running in Multiple Seats (2)	Number of Previous Runner-Ups (3)	Number of Other Previous Candidates (4)
$Post \times NewVoters$	0.105***	-0.009	-0.137***	-0.126***
	(0.037)	(0.006)	(0.038)	(0.041)
$Post \times ResChange$	-0.283***	-0.009	-0.514***	-0.371***
	(0.043)	(0.009)	(0.047)	(0.056)
Observations	6,470	6,470	6,470	6,470
R-squared	0.557	0.532	0.682	0.710
Avg Dep Variable	0.648	0.009	0.640	0.503

Notes: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting. "Other Previous Candidates" refers to candidates who placed anywhere from third place to last place in previous elections.

6.1.2 Identification Assumptions

Figure 1 provides evidence in support of the parallel trends assumption for the effect of the fraction of new voters as well as reservation status changes on political candidacy. Before redistricting occurred, constituencies with high and low proportions of new voters had the same number of incumbents nominated by major parties. Following redistricting, constituencies with a high proportion of new voters saw increases that are only present in the first election after redistricting (Figure 1). There is evidence of the parallel trends assumption holding for the effect of new voters on women's candidacy, religious minority candidacy, and independent candidates.[22] New voters only changed the nomination of incumbents in

[22]Event studies for the number of independent men and women running are presented in Appendix Figure A.6

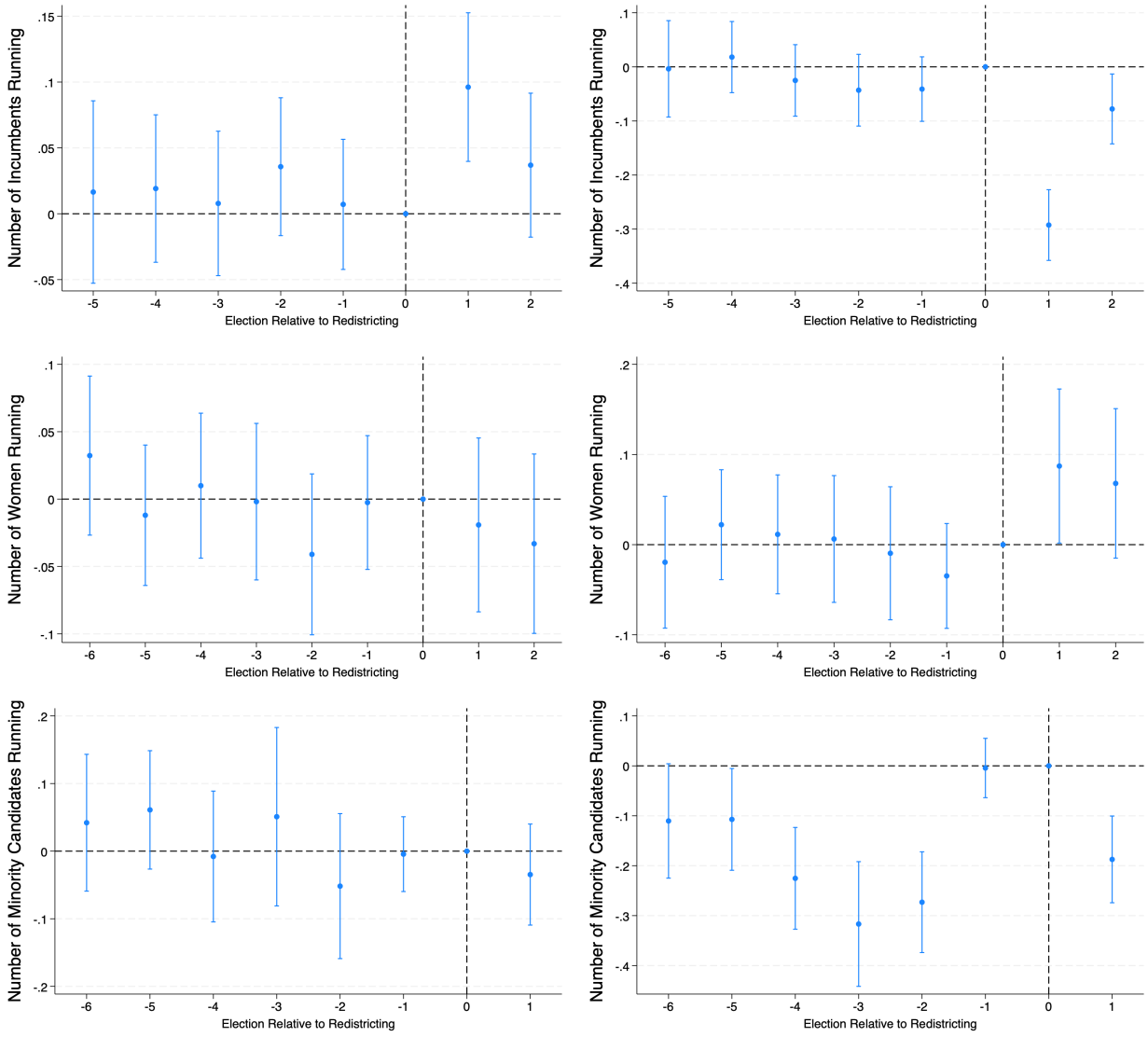
the first election following redistricting. Constituencies that experienced a change in their reservation status had the same number of incumbents and women nominated by major political parties prior to redistricting (Figure 1). The effect on incumbents is persistent in the two elections following redistricting. However, the effect in the second election after new maps were implemented is smaller than the effect immediately after redistricting.

There are potential violations of the parallel trends assumption for the effect of a reservation status change on religious minority candidacy shown in Figure 1. I therefore conduct an event study sensitivity analysis proposed by Rambachan and Roth (2023) in Appendix Figures A.7 and A.8 for incumbents and Muslim candidates, respectively. The breakdown value for a significant effect on both the number of religious minority candidates running for a major party is 0.5. This means the result is robust to allowing for violations of the parallel trends assumption up to half of the maximum violation in the pre-redistricting period (Appendix Figure A.8). Thus, the result for this outcome is not robust to violations of the parallel trends assumption. I, therefore, interpret this results cautiously.

A remaining concern for identification is that electoral redistricting in one constituency mechanically changes the bordering constituency. A constituency with a low proportion of new voters may still experience spillover effects if nearby constituencies with a high fraction of new voters see candidates or party resources shift across the boundary. Such interference would violate the Stable Unit Treatment Value Assumption (SUTVA), because the potential outcomes of one constituency would then depend not only on its own treatment status but also on the treatment status of its neighboring constituency. If a SUTVA violation were driving the results, one would expect the estimated treatment effect to decay with distance from a redrawn boundary, because constituencies further from treated units are progressively less exposed to any spillovers in candidates and party resources.

To address this concern, I compute the distance from the centroid of a constituency to the nearest "treated" constituency as defined by $NewVoters$. The median distance to the nearest "treated" constituency is 26 kilometers. Appendix Table A.5 reports a "donut" specification across increasing exclusion radii. For major party incumbents, the point estimate on $NewVoters$ remains stable, ranging from 0.043 to 0.099, as the exclusion radius increases from 0 to 20 kilometers. The effect of reservation status change is also stable. The standard errors increase modestly as the radius grows, due to the reduction in sample size as constituencies near treated units are dropped. The estimates that modestly decline are not statistically different from each other for any of the outcome variables. Due to the stability of the estimates across radii, Appendix Table A.5 supports the stable unit treatment value assumption.

Figure 1: Event Studies: Effect of Redistricting on Political Candidacy



(a) Fraction of New Voters

(b) Reservation Status Change

Notes: Each point in Panel A represents the coefficient of $NewVoters_c$ interacted with an election year indicator variable from Equation 2. Each point in Panel B is the same but uses $ResChange_c$. The bars represent the 95% confidence interval of the coefficient. Each coefficient is relative to the year before redistricting. The vertical line is at the last election before electoral boundaries were redrawn.

6.1.3 Robustness Checks

Appendix Table A.6 Panel A estimates the baseline political candidacy specification using the full pre-period sample rather than restricting to the single election immediately preceding redistricting. Using six pre-redistricting elections and one post-redistricting election, the effect on major party incumbents is 0.083 and significant at the one percent level. The effect of reservation status changes on the number of incumbents nominated by major parties is -0.280, and is also significant at the one percent level. These estimates as well as the effect on the number of women nominated by major parties are similar to those presented in Table 2. The religious minority candidacy result, however, is not robust to including six pre-redistricting elections. This suggests the religious minority candidacy finding is more sensitive to the number of pre-redistricting elections included in the analysis than the other results.

I control for the population change to ensure that the changes in the number of incumbents nominated is not driven by overall population change in a constituency. Appendix Table A.6 Panel B shows that the incumbent and religious minority results are robust to controlling for a constituency's change in the population size.

New voters in a constituency may not be the only detrimental change to incumbents. Losing a portion of your voting base may also change candidacy decision. Appendix Table A.6 shows that the results are robust controlling for the percent decline in the constituency's population. This shows that losing population is not as important for candidacy as new voters in a constituency.

When matching constituencies to each other based on which post-delimitation constituency contains the largest proportion of the pre-delimitation constituency, some post-delimitation constituencies are matched to multiple pre-delimitation constituencies. To ensure that this matching is not driving the results. I only keep the one pre-delimitation constituency with the largest proportion of voters in the post-delimitation constituency. Appendix Table A.6 Panel D shows that the results are robust to utilizing a one-to-one mapping of constituencies.

Appendix Table A.7 checks sensitivity to the construction of the $NewVoters$ variable. While the results in Table 2 use an indicator for whether a constituency's fraction of new voters is at or above the national median, Appendix Table A.7 considers five alternatives. Using a continuous measure of the fraction of new voters yields a statistically significant positive effect on major party incumbents. This results are consistent with the sign and significance pattern of the binary treatment. Restricting the comparison to the fourth quartile of fraction of new voters versus the rest of the sample produces a larger point estimate for incumbents. Comparing the first quartile to the rest of the sample shows no effect on

incumbents. Thus, the threshold for the fraction of new voters to have an effect on incumbents is the median. I also drop the top 1% of constituencies by the fraction of new voters to address concerns about outliers driving the results. These estimates are essentially the same as in Table 2 (Appendix Table A.7). Using demographic change instead of the fraction of new voters as the treatment measure, also shows that incumbents are more likely to be nominated. The effect of new voters on religious minority candidacy is not robust to these alternative specifications, creating further reason to interpret this effect with caution.

The main analysis in Table 2 clusters standard errors at the electoral constituency level. However, candidates may have run in a different constituency within the same district due to redistricting. Clustering at the constituency level could therefore understate the true standard errors if candidacy and voting behavior are correlated across constituencies within the same district. I therefore cluster at the district level in Appendix Table A.8. Beyond district level clustering, Appendix Table A.8 also reports standard errors allowing for arbitrary spatial correlation across nearby constituencies following the approach in Colella et al. (2023). Redistricting in one constituency mechanically affects the boundaries, voter composition, and candidate pools of adjacent constituencies. Thus, treatment effects and unobserved shocks may not be independent across nearby constituencies even if they fall in different districts. I report these arbitrary spatial correlation standard errors at three radii (50, 100, and 250 kilometers) to check whether effects are sensitive to spatial spillovers. Across all four sets of standard errors, the standard errors and thus significance levels for the effect of new voters and reservation status change are relatively unchanged from those presented in Table 2. Thus, the effects in Table 2 are not sensitive to clustering choice or to allowing for spatial correlation.

Appendix Table A.6, A.7, and A.8 show that the effects of new voters and changes in reservation status on major party incumbent and women nominations are robust to the choice of estimation sample, the functional form and construction of the $NewVoters$ variable, and spatial correlation. The religious minority candidacy result is an exception. The effect is robust alternative standard error specifications; however, the effect on religious minority candidacy is sensitive to to alternative constructions of $NewVoters$ and including additional pre-redistricting elections.

6.2 Voter Behavior and Political Representation

Electoral redistricting not only changes who runs for office but also who wins. Incumbents are 5.2 percentage points more likely to win reelection in a constituency with a high proportion of new voters, although the effect is not significant at conventional levels (Table

4). This is a 15.5% increase in the probability that any incumbent wins reelection. This shows that there is a one-to-one correspondence between candidacy and winning and election. While this effect is insignificant, I can rule out effects larger than 15.4 percentage points. When there is an increase in candidacy, there is an approximately equal effect on the probability of winning. New voters did not affect the number of women, religious minorities, or independents running, and it did not have an effect on the probability of these candidates winning.

Altering the reservation status of a constituency does not affect voter turnout (Table 4). There is an insignificant decrease of 1.1 percentage points. Even if this effect was significant, the magnitude is incredibly small. This is a 1.5% decline in voter turnout. The average vote margin is 11.1 percentage points. If every voter that did not turn out as a result of redistricting voted for a single politician, it would not change who won. This is corroborated by the fact that there is no effect on the vote margin (Appendix Table A.9). While the average effect on voter turnout is negative, changing a constituency from unreserved to reserved and reserved to unreserved have opposite effects. When a constituency becomes reserved, voter turnout decreases by 3.1 percentage points (Appendix Table A.10). On the other hand, voter turnout increases by 2.8 percentage points when a constituency becomes unreserved (Appendix Table A.10). These effects are still small in magnitude and still would not change the outcome of the election.

Incumbents are 14.7 percentage points more likely to lose reelection when the reservation status of their constituency changes (Table 4). Thus, incumbents are 43.8% less likely to win. This is approximately a one-to-one relationship between candidacy and winning. Fewer incumbents run, so fewer incumbents win reelection. TWhen a constituency becomes reserved, there is a larger impact on the probability an incumbent wins (Appendix Table A.10).

The probabilities that women and religious minority candidates win are differentially affected by reservation status. Fewer religious minority candidates are nominated, which translates to a 2.4 percentage point decrease in the probability a religious minority candidate wins (Table 4). This 47.1% decline in the probability a religious minority candidate wins. Similar to incumbents, there is almost a one-to-one relationship between the number of religious minority candidates running and the probability a religious minority candidate wins. The effect is completely concentrated in constituencies that became reserved (Appendix Table A.10). On the other hand, women are more likely to lose the election in constituencies that became unreserved (Appendix Table A.10). There is no change in the probability that an independent candidate wins the election.

Table 4: Effect of Electoral Redistricting on Voter Behavior and Political Representation

	Voter Turnout (1)	An Incumbent Won (2)	A Woman Won (3)	A Muslim Candidate Won (4)	An Independent Won (5)
$Post \times NewVoters$	0.007	0.052	0.021	0.011	-0.009
	(0.007)	(0.032)	(0.018)	(0.013)	(0.012)
$Post \times ResChange$	-0.011	-0.147***	-0.007	-0.024*	0.000
	(0.007)	(0.039)	(0.023)	(0.014)	(0.014)
Observations	6,470	6,470	6,470	6,470	6,470
R-squared	0.860	0.551	0.580	0.678	0.524
Avg Dep Variable	0.710	0.336	0.070	0.051	0.038

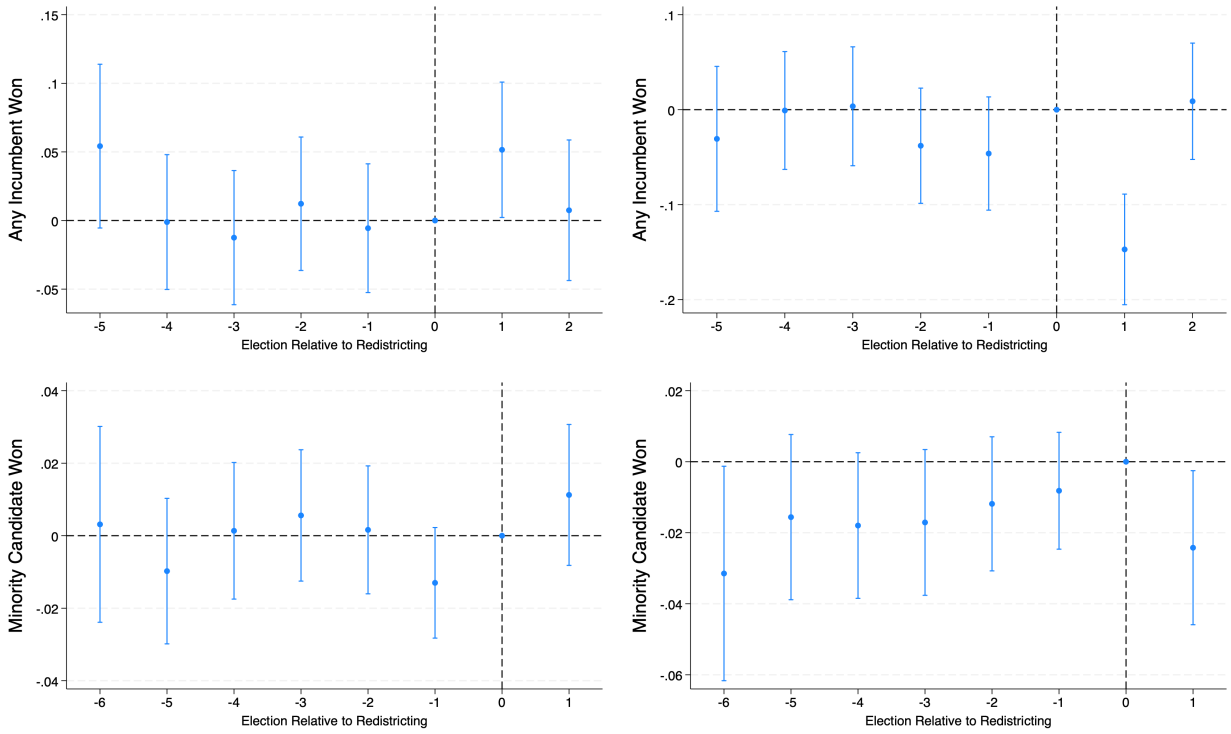
Notes: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

6.2.1 Robustness Checks

Figure 2 provides evidence in support of the parallel trends assumption for the effect of the fraction of new voters as well as reservation status changes on voter behavior and political representation. Before redistricting occurred, incumbents had the same probability of winning in constituencies with high and low proportions of new voters. Following redistricting, high-proportion constituencies saw increases that are only present in the first election with new maps (Figure 2). The effect of reservation status changes on incumbents winning is also just observed in the first election post-delimitation (Figure 2). I conduct an event study sensitivity analysis for the effect of reservation status change on the probability an incumbent won in Appendix Figure A.10. The breakdown value is 2, which means that the results are robust to violations up to twice as large as those seen pre-redistricting. There is evidence of the parallel trends assumption holding for the effect of new voters and reservation status changes on religious minority candidates winning as well. Event studies for the effects of redistricting on voter turnout, women winning, and independents winning are presented in Appendix Figure A.9.

To test for SUTVA violations, I conduct another "donut" specification in Appendix Table A.11. The effect of new voters on voter turnout remains stable, ranging from 0.007 to 0.011, as the exclusion radius increases from 0 to 20 kilometers. Similarly, the estimate for the effect of reservation status change on any incumbent winning is robust, ranging from

Figure 2: Event Studies: Effect of Redistricting on Voter Behavior and Representation



(a) Fraction of New Voters

(b) Reservation Status Change

Notes: Each point in Panel A represents the coefficient of $NewVoters_c$ interacted with an election year indicator variable from Equation 2. Each point in Panel B is the same but uses $ResChange_c$. The bars represent the 95% confidence interval of the coefficient. Each coefficient is relative to the year before redistricting. The vertical line is at the last election before electoral boundaries were redrawn.

-0.144 to -0.181. The effect of reservation status changes on religious minorities shows the same pattern, with the effect on the ranging from -0.018 to -0.026. Standard errors increase modestly as the radius grows due to the decreasing sample size. The estimates that modestly decline are not statistically different from each other for any of the outcome variables. Due to the stability of the estimates across radii, Appendix Table A.11 supports the stable unit treatment value assumption.

Appendix Table A.12 Panel A re-estimates equation 1 using all six elections before redistricting. The effect of reservation status changes on voter turnout and the probability any incumbent wins are robust to this specification. The probability an incumbent wins due to new voters increases by 4.8 percentage points and is significant at the 5% level. The effect on the probability a religious minority candidate wins is not robust to this sample. The Table 4 coefficient of -0.024 shrinks to -0.011 and loses significance. I therefore continue to cautiously interpret the effect of redistricting on religious minority candidates.

Appendix Table A.12 Panel B shows that the probability an incumbent wins reelection is not a result of overall population size changes. Panel C indicates that losing population in a constituency does not change the effects of new voters or reservation status changes on the probability an incumbent wins (Appendix Table A.12). Whereas, Panel D shows that the one-to-one mapping only makes the results stronger (Appendix Table A.12).

Appendix Table A.13 checks sensitivity to the construction of $NewVoters$. The continuous, fourth quartile, dropping top 1%, and demographic change specifications all preserve the sign and significance of the effect of redistricting on the probability an incumbent wins. Thus, one should not be concerned that the definition of $NewVoters$ is driving the results.

Appendix Table A.14 reports district-level clustering as well as standard errors allowing for arbitrary spatial correlation at 50, 100, and 250 kilometers. The results are robust to using all four of these alternative standard errors, indicating the results is not sensitive to clustering choice.

Together, Appendix Tables A.12, A.13, and A.14 show that the negative effect of redistricting on the probability an incumbent wins is robust to the choice of estimation sample, the construction of the treatment variable, and alternative choices of standard errors. As with the political candidacy results, the effect on religious minority candidate's electoral outcomes are not robust to including additional pre-redistricting elections.

7 Does Redistricting Shape Development Outcomes?

Now, I examine how redistricting alters policy and development outcomes. Given that prior literature demonstrates that entrenched politicians and dynastic politics are detrimental

for economic development (Acemoglu and Robinson, 2006; George, 2024), and new leaders are more responsive and incorporate citizen needs into policy (Bazzi et al., 2025), a key question is whether the redistricting-induced decline in incumbents translates into tangible welfare gains, losses, or both.

Having a high proportion of new voters in a constituency slowed the growth of primary schools. The provision of primary schools decreases by 1.2 percentage points relative to places with a low proportion of new voters (Table 5). The effect is not destroying primary schools, but rather slowed growth. This slower growth is consistent with the research that finds incumbents are bad for growth. The fraction of new voters did not have a significant effect on changing representation. Thus, having similar representation as before redistricting slows the provision of public goods. The fraction of new voters in a constituency does not alter the provision of other public goods, including middle schools, health facilities, and paved roads, and does not change nighttime light growth.

On the other hand, changing the reservation status of a constituency increases the provision of middle schools, primary health centers, and nighttime light growth. The provision of middle schools increases by 1.71 percentage points, which is a 4.4% increase (Table 5). This effect on middle schools accounts for 13.3% of the 12.8 percentage point increase between 2001 and 2011. Changing the reservation status of a constituency increases the provision of primary health centers by 0.9 percentage points (Table 5). This is an 11.7% rise in primary health centers. Nighttime lights grow by 1.23 percentage points when a constituency's reservation status changes.[23] Nighttime lights grow annually by 7.5 percentage points on average, so redistricting increases growth by 16.4%. This effect on nighttime lights is 32% as large as the effect of programs specifically targeting electrification in India (Burlig and Preonas, 2024).

These results are consistent with the representation channel. Having a high proportion of new voters in a constituency does not necessarily change the social identity of the winning candidate. The officeholder's underlying policy preferences are more likely to persist even as the electorate around her shifts. A change in reservation status, by contrast, forces a change in the identity of who holds the seat. Politicians from different social groups have been shown to bring systematically different policy priorities to office (Chattopadhyay and Duflo, 2004; Pande, 2003). Thus, reservation status changes generate a larger effect on economic development than the fraction of new voters because only the reservation status changes require a change to the identity of the officeholder.

The constituencies that see the sharpest disruption to incumbency see gains in the provi-

[23] Converting the estimate from the Poisson pseudo maximum likelihood estimation to percentage points uses the following formula: $100 \times (e^{0.01\hat{\beta}} - 1)$.

sion of middle schools, primary health centers, and nighttime light growth. This is consistent with the literature showing that entrenched incumbent politicians are detrimental for economic growth. When redistricting changes who wins, public goods provision and nighttime lights grow. Thus, the loss of incumbency is one mechanism through which redistricting affects policy and economic development. Appendix Table A.15 shows the gains in public goods provision are distributed between both constituencies that became reserved and those that became unreserved, although along different margins. Constituencies that became reserved see gains in primary health center provision and nighttime light growth. Additionally, constituencies that became unreserved had faster growth in the provision of primary schools, middle schools, and paved roads. The growth in nighttime lights is observed in constituencies that became reserved (Appendix Table A.15). The effect on luminosity growth in constituencies that became unreserved is positive and statistically the same as the effect in constituencies that became reserved. These results show that new politicians increase the provision of public goods as well as nighttime light growth. This suggest redistricting is altering policy through representation. I do not have data on voter preferences, so I cannot rule out changes in voter preferences also contributing to the change in public goods provision. Thus, both changes in representation paired with different voter preferences alter the provision of public goods. [24]

The estimated 1.23 percentage point increase in nighttime light growth suggests a $2.26 billion annual gain in localized economic activity during the five-year legislative term following redistricting. A one percentage point increase in nighttime lights is associated with a 0.15 percentage point increase in GDP growth (Baskaran et al., 2024). Using Baskaran et al. (2024)'s estimate, a 1.23 point increase in nighttime light growth corresponds to a 0.1845 increase in India's GDP growth rate. Redistricting went into effect in 2008. In 2007, India's GDP was $1.22 trillion. Redistricting's 0.1845 percentage point increase in GDP growth indicates there was a $2.26 billion gain in the economy as a result of redistricting. Nighttime lights serve as a valuable proxy for subnational economic activity where gross domestic product is unobserved, and mapping these luminosity changes directly to aggregate welfare losses has underlying assumptions. Applying elasticities to translate this luminosity decline into a specific monetary value suggests an expansion in localized economic activity. Thus, the disruption of local political representation has benefits for local economic development.

I now conduct several robustness to show the results are not driven by the sample selec-

[24]The third channel though which redistricting could affect economic development is through political competition. Using the usual measure of political competition, including vote number of candidates running, number of parties contesting, and vote margin, redistricting does not alter these measures (Appendix Tables A.4 and A.9). As there is not an increase in political competition, this channel would not be the reason policies change.

Table 5: Effect of Electoral Redistricting on Development Outcomes

	Primary Schools (1)	Middle Schools (2)	Health Centers (3)	Health Subcenters (4)	Paved Roads (5)	Luminosity Growth (6)
$Post \times NewVoters$	-1.226*	-1.249	-0.586	0.097	0.852	-0.016
	(0.682)	(0.939)	(0.413)	(0.866)	(1.031)	(0.344)
$Post \times ResChange$	0.160	1.709*	0.904**	-0.063	1.442	1.221***
	(0.822)	(1.038)	(0.451)	(0.913)	(1.195)	(0.401)
Observations	4,872	4,872	4,872	4,872	4,872	5,680
R-squared	0.705	0.820	0.875	0.809	0.784	0.020
Avg Dep Variable	87.251	39.112	7.702	22.317	68.635	7.484

Notes: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. The R-squared value in Column 6 is a pseudo-R-squared value from the PPML estimation.

tion, SUTVA violations, or spatial correlation. To show the results are not driven by sample composition, Appendix Table A.16 estimates the candidacy and representation results on the sample used for the development analysis. The SHRUG data has fewer constituencies than the overall election data, so this analysis will ensure that the null results for the effect of new voters on development are not due to the sample. Restricting to the public goods sample, the effects of redistricting on incumbent candidacy and electoral outcomes are similar to those found in Tables 2 and 4, indicating that the null development results in Table 5 are not a result of the sample.

To test for SUTVA violations, Appendix Table A.17 reports the "donut" specification across exclusion radii of 0 to 20 kilometers for all four development outcomes. The effect of reservation status changes on nighttime light growth declines somewhat as the radius increases (from 0.012 at the full sample to 0.005 at 20km), while the middle school and primary health center coefficients remains stable and significant throughout. For both of these outcomes, the main specification estimate is not statistically different than the estimate with an exclusion radius of 20 kilometers. Thus, Appendix Table A.17 supports the stable unit treatment value assumption.

Appendix Table A.18 estimates the effect of redistricting on development outcomes using the 1991, 2001, and 2011 population censuses. The results are robust to the inclusion of

the 1991 census. The effects of reservation status change on middle schools and paved roads are stronger in this specification. Panel B shows that the nighttime lights and middle schools results are robust to controlling for population change. This shows that the results are not driven by equalizing spending per capita on constituents. Panel C indicates that a constituency's population shrinking are not driving the results. One may expect public goods provision to slow when population shrinks due to economies of scale. As the fraction of new voters in a constituency and population shrinkage are strongly correlated, controlling for population shrinkage strengthens the results. This further shows that incumbents slow economic development. The one-to-one matching of constituencies in Panel D shows the results are robust to this sample as well.

Appendix Table A.19 determines if the results are due to the constructions of the $NewVoters$ variable. The continuous and fourth quartile measures both yield a significant negative effect on middle schools, but yield a similar result for primary schools provision. Appendix Table A.20 reports district-level clustering as well as standard errors allowing for spatial correlation at 50, 100, and 250 kilometers. The effect on paved roads and nighttime light growth retains significance across all different clustering specifications. Appendix Table A.21 addresses concerns about low sensitivity and saturation in the nighttime light growth outcome due to satellite sensors by dropping the top and bottom one percent of the sample. The results are robust to this specification; thus one should not be concerned about the technical limitations of the nighttime light data.

8 Conclusion

This paper examines how electoral redistricting reshapes who runs for office, who wins, and economic development. Redistricting operates through two channels: uncertainty and legal restrictions on candidacy. Uncertainty forces political parties to change their electoral strategy. Parties nominate more incumbents and fewer previous runners-up. Legal restrictions reduce the number of incumbents nominated by major parties. While parties do not adjust their nomination of women or religious minority candidates in response to new voters, party nomination strategies respond to a reservation status change. Parties increase the number of women nominated when a constituency's becomes reserved, while religious minority nominations fall almost completely, reflecting parties' legal inability to nominate certain candidates as well as trade offs in the spillovers of affirmative action.

These candidacy effects carry through to electoral outcomes. Incumbents are 43.8% less likely to win when their constituency's reservation status changes. The loss of incumbency as well as changes in voter preferences generate economic gains. Constituencies experiencing a

reservation status change see a 4.4% increase in middle school provision and a 11.7% increase in the provision of primary health centers. These same constituencies have a 1.23 percentage point increase in nighttime light growth. The effect on public goods provision is concentrated in constituencies that became unreserved, while nighttime light growth is concentrated in constituencies that became reserved. These overall results are consistent with prior evidence that entrenched incumbents are detrimental to local economic development, and that new leadership is more responsive to constituent needs.

These findings carry implications for democracies that use redistricting to redraw electoral maps. Redistricting is treated as a technical exercise that keeps district populations equal or as a way to solidify electoral majorities. This paper shows that, even when an independent body redraws boundaries, electoral redistricting alters who runs, who wins, and what constituents receive. Yet independence of the map drawing process does not guarantee independence from the strategic responses of political parties. Faced with unfamiliar electorates, parties fall back on their incumbents, using nomination strategy to preserve reelection prospects that new boundaries might otherwise disrupt. Only where parties' room to maneuver is constrained, as when changes in affirmative action status legally bar incumbents from running, does redistricting produce meaningful turnover. Then, it is in these constituencies that the economic gains emerge. Fair maps, thus, are not sufficient to unseat entrenched incumbents. Policymakers and advocates for independent commissions should recognize that parties adapt to the boundaries they are given. Taking map-drawing out of politicians' hands may not be enough on its own; fair representation and development gains of redistricting also depend on limiting political parties' ability to offset it.

References

Acemoglu, Daron, and James A. Robinson. 2006. "Economic Backwardness in Political Perspective." *American Political Science Review*, 100(1): 115–131.

Afridi, Farzana, Vegard Iversen, and M. R. Sharan. 2017. "Women Political Leaders, Corruption, and Learning: Evidence from a Large Public Program in India." *Economic Development and Cultural Change*, 66(1): 1–30.

Akhtari, Mitra, Diana Moreira, and Laura Trucco. 2022. "Political Turnover, Bureaucratic Turnover, and the Quality of Public Services." *American Economic Review*, 112(2): 442–493.

Alesina, Alberto, William Easterly, and Janina Matuszeski. 2011. "Artificial States." *Journal of the European Economic Association*, 9(2): 246–277.

Ansolabehere, Stephen, Alan Gerber, and Jim Snyder. 2002. "Equal Votes, Equal Money: Court-Ordered Redistricting and Public Expenditures in the American States." *American Political Science Review*, 96(4): 767–777.

Anzia, Sarah F., and Christopher R. Berry. 2011. "The Jackie (and Jill) Robinson Effect: Why Do Congresswomen Outperform Congressmen?" *American Journal of Political Science*, 55(3): 478–493. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1540-5907.2011.00512.x.

Ardanaz, Martin, and Carlos Scartascini. 2013. "Inequality and Personal Income Taxation: The Origins and Effects of Legislative Malapportionment." *Comparative Political Studies*, 46(12): 1636–1663.

Arora, Ashna. 2022. "Election by Community Consensus: Effects on Political Selection and Governance." *The Review of Economics and Statistics*, 104(2): 321–335.

Asher, Sam, Tobias Lunt, Ryu Matsuura, and Paul Novosad. 2021. "Development Research at High Geographic Resolution: An Analysis of Night-Lights, Firms, and Poverty in India Using the SHRUG Open Data Platform." *World Bank Economic Review*, 35(4): 845–871.

Ashworth, John, Benny Geys, Bruno Heyndels, and Fanny Wille. 2014. "Competition in the political arena and local government performance." *Applied Economics*, 46(19): 2264–2276.

Ashworth, Scott. 2006. "Campaign Finance and Voter Welfare with Entrenched Incumbents." *American Political Science Review*, 100(1): 55–68.

Baker, Andrew C., David F. Larcker, and Charles C. Y. Wang. 2022. "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics*, 144(2): 370–395.

Baskaran, Thushyanthan, Brian Min, and Yogesh Uppal. 2015. "Election cycles and electricity provision: Evidence from a quasi-experiment with Indian special elections." *Journal of Public Economics*, 126: 64–73.

Baskaran, Thushyanthan, Sonia Bhalotra, Brian Min, and Yogesh Uppal. 2024. "Women legislators and economic performance." *Journal of Economic Growth*, 29(2): 151–214.

Bauer, Nichole M., and Ivy A. M. Cargile. 2023. "Women Get the Job Done: Differences in Constituent Communication from Female and Male Lawmakers." *Politics & Gender*, 19(4): 1110–1133.

Bazzi, Samuel, Masyhur Hilmy, Benjamin Marx, Mahvish Shaukat, and Andreas Stegmann. 2025. "It Takes a Village Election: Turnover and Performance in Local Bureaucracies." *CEPR Discussion Papers*. Number: DP19994.

Beath, Andrew, Fotini Christia, Georgy Egorov, and Ruben Enikolopov. 2016. "Electoral Rules and Political Selection: Theory and Evidence from a Field Experiment in Afghanistan." *The Review of Economic Studies*, 83(3): 932–968.

Bertrand, Marianne, Rema Hanna, and Sendhil Mullainathan. 2010. "Affirmative action in education: Evidence from engineering college admissions in India." *Journal of Public Economics*, 94(1): 16–29.

Besley, Timothy, and Stephen Coate. 1997. "An Economic Model of Representative Democracy." *The Quarterly Journal of Economics*, 112(1): 85–114.

Besley, Timothy, Torsten Persson, and Daniel M. Sturm. 2010. "Political Competition, Policy and Growth: Theory and Evidence from the US." *The Review of Economic Studies*, 77(4): 1329–1352.

Bhalotra, Sonia, Irma Clots-Figueras, and Lakshmi Iyer. 2018. "Pathbreakers? Women's Electoral Success and Future Political Participation." *The Economic Journal*, 128(613): 1844–1878.

Bhavnani, Rikhil R. 2017. "Do the Effects of Temporary Ethnic Group Quotas Persist? Evidence from India." *American Economic Journal: Applied Economics*, 9(3): 105–123.

Bhavnani, Rikhil R. 2018. "The Effects of Malapportionment on Cabinet Inclusion: Subnational Evidence from India." *British Journal of Political Science*, 48(1): 69–89.

Bhavnani, Rikhil R. 2021. "The effects of malapportionment on economic development." *PLOS ONE*, 16(12): e0259150.

Bluhm, Richard, Roland Hodler, and Paul Schaudt. 2024. "Ethnofederalism: Subnational borders and the salience of ethnicity in politics." *Working Paper*.

Burgess, Robin, Michael Greenstone, Nicholas Ryan, and Anant Sudarshan. 2020. "The Consequences of Treating Electricity as a Right." *Journal of Economic Perspectives*, 34(1): 145–169.

Burlig, Fiona, and Louis Preonas. 2024. "Out of the Darkness and into the Light? Development Effects of Rural Electrification." *Journal of Political Economy*, 132(9): 2937–2971.

Caetano, Carolina, Brantly Callaway, Stroud Payne, and Hugo Sant'Anna Rodrigues. 2024. "Difference in Differences with Time-Varying Covariates." arXiv:2202.02903 [econ].

Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with multiple time periods." *Journal of Econometrics*, 225(2): 200–230.

Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2025. "Difference-in-differences with a Continuous Treatment." *Working Paper*.

Cassan, Guilhem. 2019. "Affirmative action, education and gender: Evidence from India." *Journal of Development Economics*, 136: 51–70.

Cassan, Guilhem, and Lore Vandewalle. 2021. "Identities and public policies: Unexpected effects of political reservations for women in India." *World Development*, 143: 105408.

Cassidy, Traviss, and Tejaswi Velayudhan. 2026. "Government Fragmentation and Economic Growth." *The Review of Economics and Statistics*, 108(2): 372–389.

Chand, T. R. Kiran, K. V. S. Badarinath, C. D. Elvidge, and B. T. Tuttle. 2009. "Spatial characterization of electrical power consumption patterns over India using temporal DMSP-OLS night-time satellite data." *International Journal of Remote Sensing*, 30(3): 647–661. _eprint: https://doi.org/10.1080/01431160802345685.

Chattopadhyay, Raghabendra, and Esther Duflo. 2004. "Women as Policy Makers: Evidence from a Randomized Policy Experiment in India." *Econometrica*, 72(5): 1409–1443.

Clayton, Amanda. 2021. "How Do Electoral Gender Quotas Affect Policy?" *Annual Review of Political Science*, 24(1): 235–252. _eprint: https://doi.org/10.1146/annurev-polisci-041719-102019.

Clots-Figueras, Irma. 2012. "Are Female Leaders Good for Education? Evidence from India." *American Economic Journal: Applied Economics*, 4(1): 212–244.

Colella, Fabrizio, Rafael Lalive, Seyhun Orcan Sakalli, and Mathias Thoenig. 2023. "acreg: Arbitrary correlation regression." *The Stata Journal*, 23(1): 119–147.

DeSilver, Drew. 2025. "U.S. stands out globally in how it draws legislative districts." *Pew Research Center*.

Dinkelman, Taryn. 2011. "The Effects of Rural Electrification on Employment: New Evidence from South Africa." *American Economic Review*, 101(7): 3078–3108.

Downs, Anthony. 1957. "An Economic Theory of Political Action in a Democracy." *Journal of Political Economy*, 65(2): 135–150.

Erikson, Robert S. 1972. "Malapportionment, Gerrymandering, and Party Fortunes in Congressional Elections." *American Political Science Review*, 66(4): 1234–1245.

Fraga, Bernard L. 2016. "Redistricting and the Causal Impact of Race on Voter Turnout." *The Journal of Politics*, 78(1): 19–34.

Friedman, John N., and Richard T. Holden. 2008. "Optimal Gerrymandering: Sometimes Pack, but Never Crack." *American Economic Review*, 98(1): 113–144.

Fryer, Roland G., and Richard Holden. 2011. "Measuring the Compactness of Political Districting Plans." *The Journal of Law and Economics*, 54(3): 493–535.

George, Siddharth. 2024. "Like Father, Like Son? The Effect of Political Dynasties on Economic Development." *Working Paper*.

Goodman-Bacon, Andrew. 2021. "Difference-in-differences with variation in treatment timing." *Journal of Econometrics*, 225(2): 254–277.

Gul, Faruk, and Wolfgang Pesendorfer. 2010. "Strategic Redistricting." *American Economic Review*, 100(4): 1616–1641.

Gulzar, Saad. 2021. "Who Enters Politics and Why?" *Annual Review of Political Science*, 24(Volume 24, 2021): 253–275.

Hayes, Danny, and Seth C. McKee. 2009. "The Participatory Effects of Redistricting." *American Journal of Political Science*, 53(4): 1006–1023.

Henderson, J. Vernon, Adam Storeygard, and David N. Weil. 2012. "Measuring Economic Growth from Outer Space." *American Economic Review*, 102(2): 994–1028.

Henderson, Vernon, Adam Storeygard, and David N. Weil. 2011. "A Bright Idea for Measuring Economic Growth." *American Economic Review*, 101(3): 194–199.

Iyer, Lakshmi, and Coleson Weir. 2025. "The colonial legacy in India: How persistent are the effects of historical institutions?" *Journal of Development Economics*, 177: 103576.

Iyer, Lakshmi, Coleson Weir, and Susan Ostermann. 2026. "Redrawing the Lines: Did Political Incumbents Influence Electoral Redistricting in India?" *Working Paper*.

Jensenius, Francesca Refsum. 2013. "Was the Delimitation Commission Unfair to Muslims?" *Studies in Indian Politics*, 1(2): 213–229.

Karekurve-Ramachandra, Varun, and Alexander Lee. 2020. "Do Gender Quotas Hurt Less Privileged Groups? Evidence from India." *American Journal of Political Science*, 64(4): 757–772.

Kjelsrud, Anders, Kalle Moene, and Lore Vandewalle. 2024. "Political Competition Over Life and Death: Social Provision and Infant Mortality in India." *Journal of Human Resources*.

Lawless, Jennifer L. 2015. "Female Candidates and Legislators." *Annual Review of Political Science*, 18: 349–366.

Lawless, Jennifer L., and Kathryn Pearson. 2008. "The Primary Reason for Women's Underrepresentation? Reevaluating the Conventional Wisdom." *The Journal of Politics*, 70(1): 67–82.

Lazarus, Jeffrey, and Amy Steigerwalt. 2018. *Gendered Vulnerability: How Women Work Harder to Stay in Office*. University of Michigan Press.

Lipscomb, Molly, A. Mushfiq Mobarak, and Tania Barham. 2013. "Development Effects of Electrification: Evidence from the Topographic Placement of Hydropower Plants in Brazil." *American Economic Journal: Applied Economics*, 5(2): 200–231.

Lipscomb, Molly, and Ahmed Mushfiq Mobarak. 2017. "Decentralization and Pollution Spillovers: Evidence from the Re-drawing of County Borders in Brazil." *The Review of Economic Studies*, 84(1): 464–502.

Magesan, Arvind, Andrea Szabó, and Gergely Ujhelyi. 2025. "Do Parties and Voters Disagree? An Equilibrium Analysis of Candidate Selection in India." *Working Paper*.

Mahadevan, Meera. 2024. "The Price of Power: Costs of Political Corruption in Indian Electricity." *American Economic Review*, 114(10): 3314–3344.

Marx, Benjamin, Vincent Pons, and Vincent Rollet. 2025. "Electoral Turnovers." *The Review of Economic Studies*, 92(5): 3306–3339.

McKinnie, Kelly, and Erin Szalda-Petree. 2024. "Enumeration and sampling analysis of Montana's 2020 congressional redistricting map." *Journal of Computational Social Science*, 8(1): 10.

Meserve, Stephen A., Daniel Pemstein, and William T. Bernhard. 2020. "Gender, Incumbency and Party List Nominations." *British Journal of Political Science*, 50(1): 1–15.

Michalopoulos, Stelios, and Elias Papaioannou. 2016. "The Long-Run Effects of the Scramble for Africa." *The American economic review*, 106(7): 1802–1848. Place: Nashville.

Min, Brian, Kwawu Mensan Gaba, Ousmane Fall Sarr, and Alassane Agalassou. 2013. "Detection of rural electrification in Africa using DMSP-OLS night lights imagery." *International Journal of Remote Sensing*, 34(22): 8118–8141. _eprint: https://doi.org/10.1080/01431161.2013.833358.

Narasimhan, Veda, and Jeffrey Weaver. 2024. "Polity Size and Local Government Performance: Evidence from India." *American Economic Review*, 114(11): 3385–3426.

Niederle, Muriel, and Lise Vesterlund. 2007. "Do Women Shy Away From Competition? Do Men Compete Too Much?*." *The Quarterly Journal of Economics*, 122(3): 1067–1101.

Norris, Pippa, and Ronald Inglehart. 2001. "Women and Democracy: Cultural Obstacles to Equal Representation." *Journal of Democracy*, 12(3): 126–140.

Osborne, Martin J., and Al Slivinski. 1996. "A Model of Political Competition with Citizen-Candidates." *The Quarterly Journal of Economics*, 111(1): 65–96.

Padovano, Fabio, and Roberto Ricciuti. 2009. "Political competition and economic performance: evidence from the Italian regions." *Public Choice*, 138(3): 263–277.

Pande, Rohini. 2003. "Can Mandated Political Representation Increase Policy Influence for Disadvantaged Minorities? Theory and Evidence from India." *American Economic Review*, 93(4): 1132–1151.

Preece, Jessica Robinson, Olga Bogach Stoddard, and Rachel Fisher. 2016. "Run, Jane, Run! Gendered Responses to Political Party Recruitment." *Political Behavior*, 38(3): 561–577.

Rambachan, Ashesh, and Jonathan Roth. 2023. "A More Credible Approach to Parallel Trends." *The Review of Economic Studies*, 90(5): 2555–2591.

Shaukat, Mahvish. 2019. "Too Close to Call: Electoral Competition and Politician Behavior in India." *Working Paper*.

Sun, Liyang, and Sarah Abraham. 2021. "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects." *Journal of Econometrics*, 225(2): 175–199.

Thomsen, Danielle M., and Bailey K. Sanders. 2020. "Gender Differences in Legislator Responsiveness." *Perspectives on Politics*, 18(4): 1017–1030.

Toral, Guillermo. 2024. "Turnover: How Lame-Duck Governments Disrupt the Bureaucracy and Service Delivery before Leaving Office." *The Journal of Politics*, 86(4): 1348–1367.

Tuttle, Benjamin T., Sharolyn Anderson, Chris Elvidge, Tilottama Ghosh, Kim Baugh, and Paul Sutton. 2014. "Aladdin's Magic Lamp: Active Target Calibration of the DMSP OLS." *Remote Sensing*, 6(12): 12708–12722.

Uppal, Yogesh. 2009. "The disadvantaged incumbents: estimating incumbency effects in Indian state legislatures." *Public Choice*, 138(1): 9–27.

Weir, Coleson. 2026*a*. "Dividing Lines: Redistricting, Race, and Party Vote Shares." *Work-in-Progress*.

Weir, Coleson. 2026*b*. "Gerrymandering Gender: Electoral Redistricting and Women's Political Representation in the United States." *Work-in-Progress*.

A Appendix: Additional Tables and Figures

Figure A.1: India Constituency Population Distribution

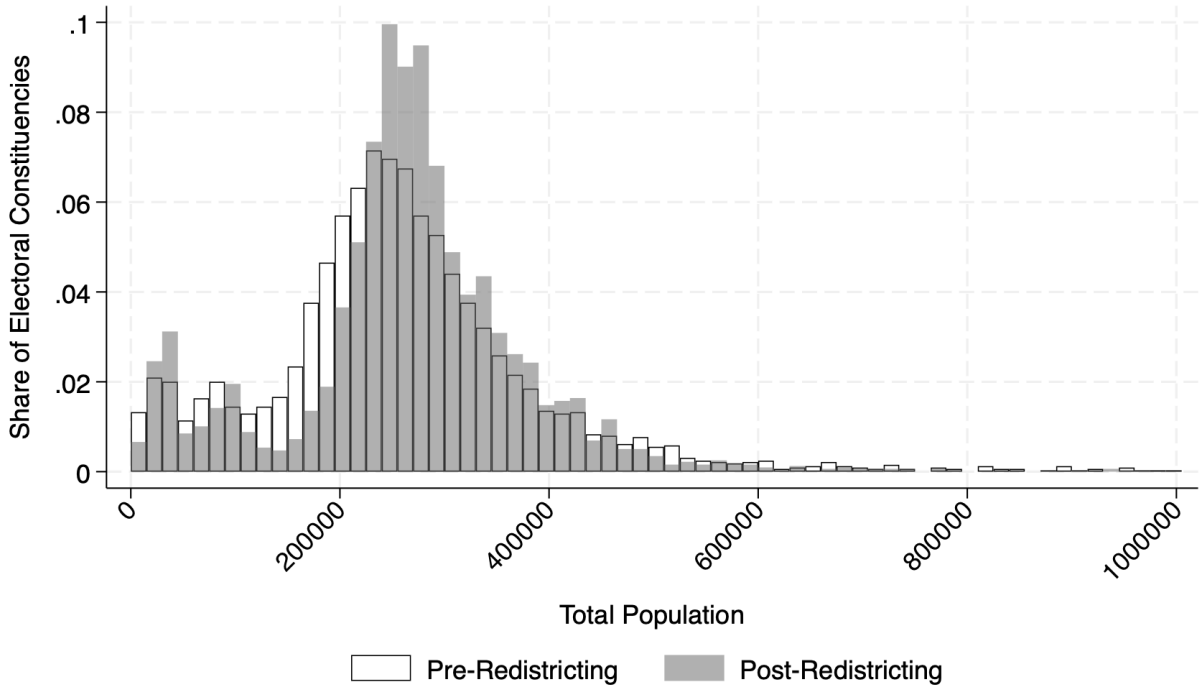


Figure A.2: Diagram of Conceptual Framework

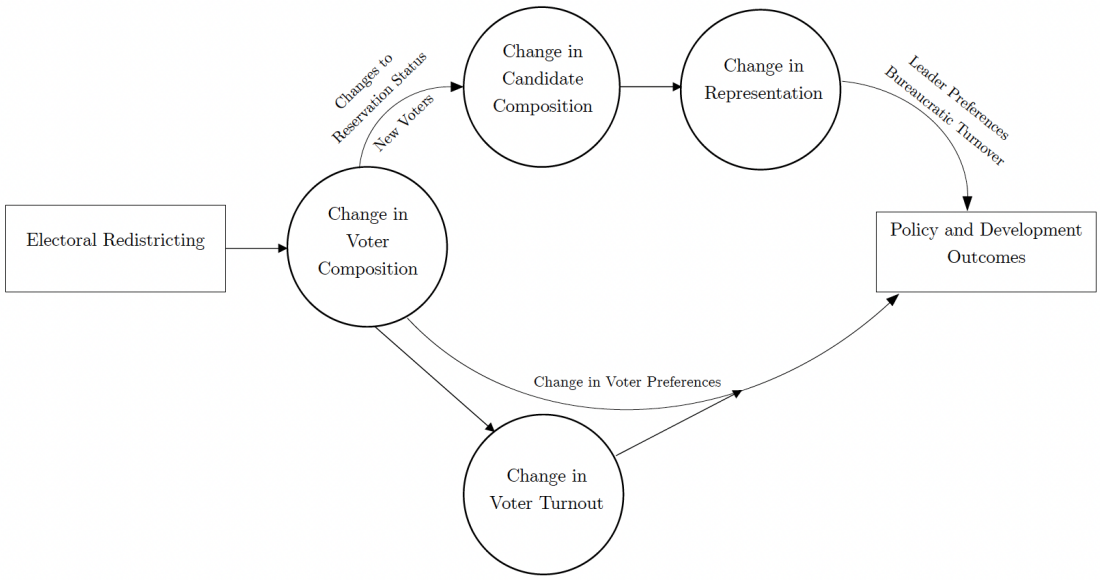


Figure A.3: Constituency Maps for Anantapur District, Andhra Pradesh, India

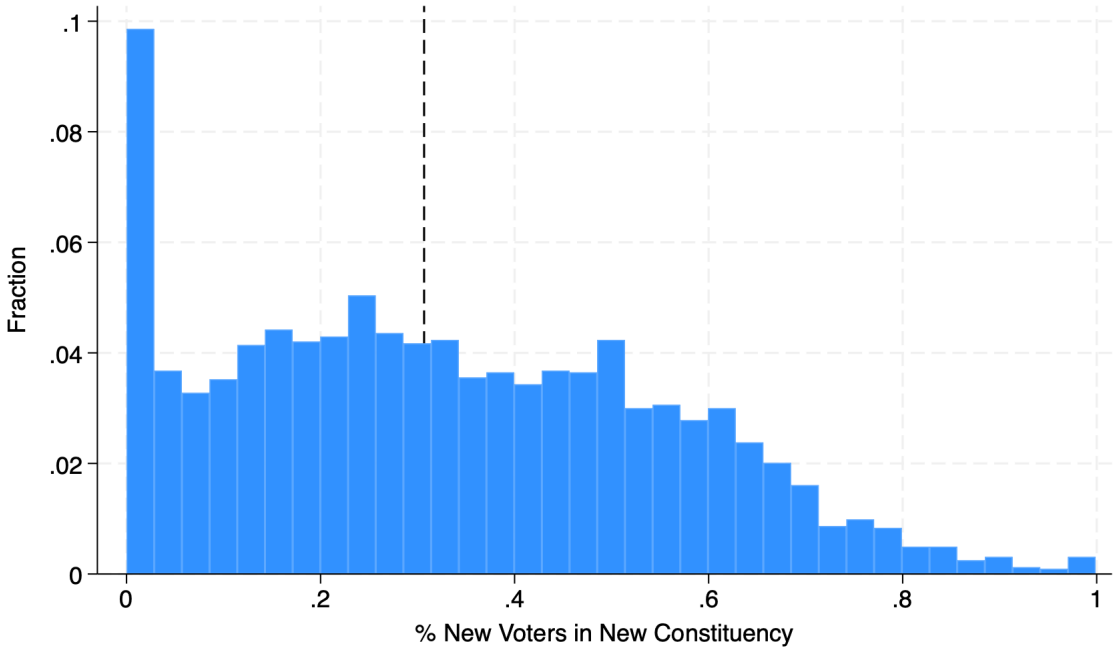


(a) Pre-Electoral Redistricting



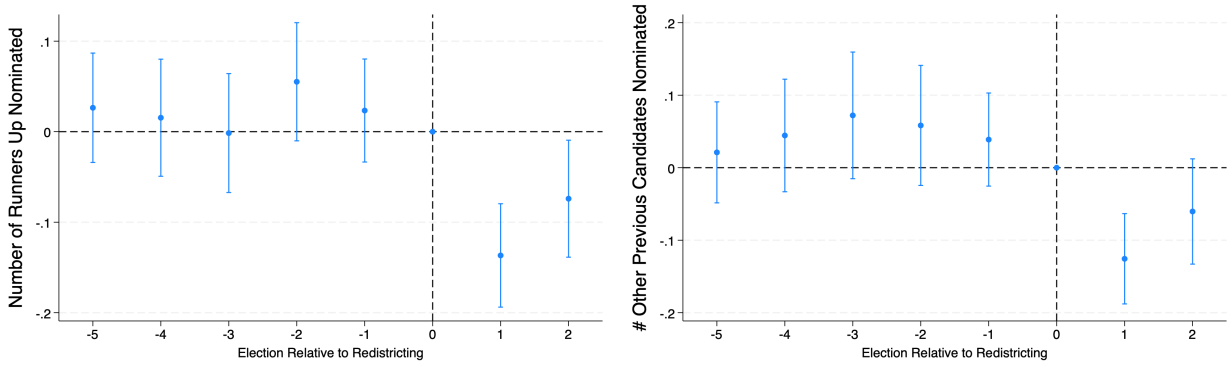
(b) Post-Electoral Redistricting

Figure A.4: Distribution of the Fraction of New Voters



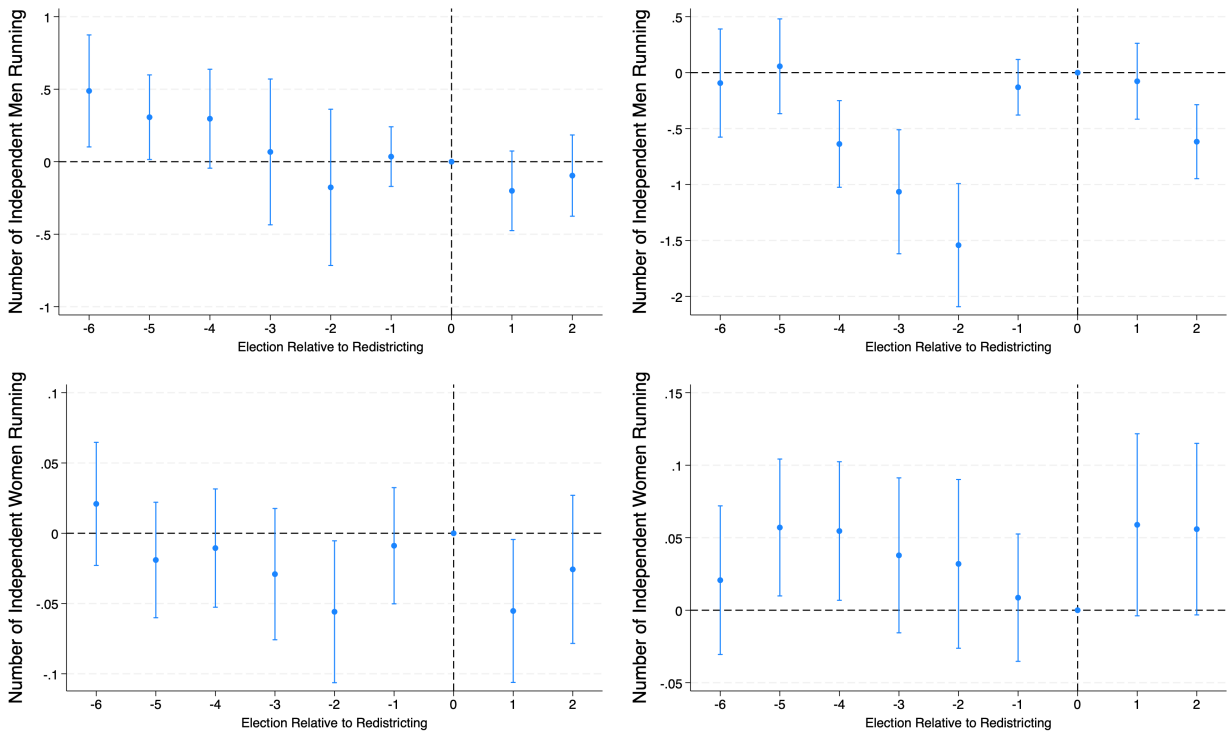
Notes: The dashed line is located at the median of the distribution.

Figure A.5: Event Studies: Effect of New Voters on Party Strategy



Notes: Each point represents the coefficient of $NewVoters_c$ interacted with an election year indicator variable from Equation 2. The bars represent the 95% confidence interval of the coefficient. Each coefficient is relative to the year before redistricting. The vertical line is at the last election before electoral boundaries were redrawn.

Figure A.6: Event Studies: Effect of Redistricting on Political Candidacy

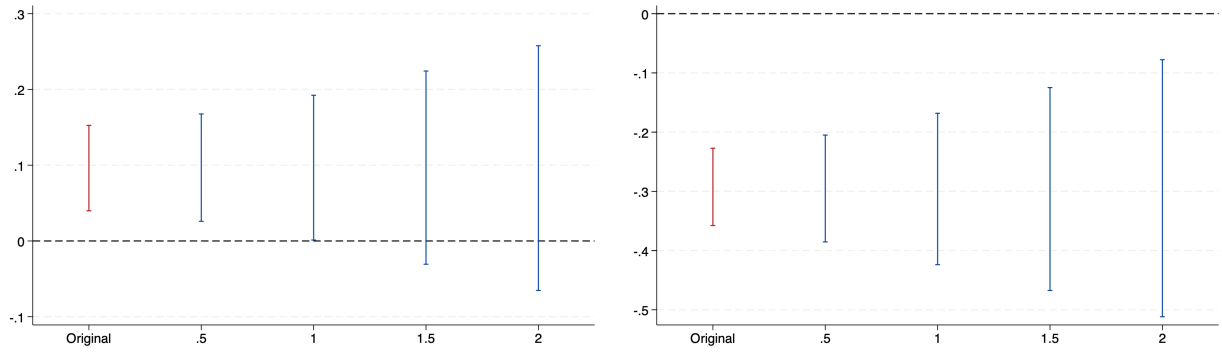


(a) Fraction of New Voters

(b) Reservation Status Change

Notes: Each point in Panel A represents the coefficient of $NewVoters_c$ interacted with an election year indicator variable from Equation 2. Each point in Panel B is the same but uses $ResChange_c$. The bars represent the 95% confidence interval of the coefficient. Each coefficient is relative to the year before redistricting. The vertical line is at the last election before electoral boundaries were redrawn.

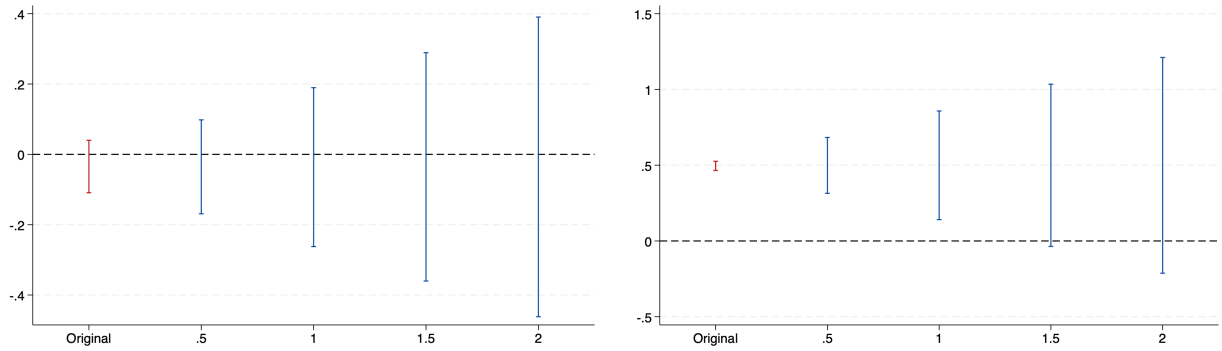
Figure A.7: Event Study Sensitivity Analysis: Effect of Redistricting on Incumbent Nominations



(a) New Voters (b) Reservation Status Change

Notes: This figure depicts the results of the event-study sensitivity analysis proposed in Rambachan and Roth (2023). Each panel plots the robust 95% confidence sets for the treatment effect in the election year after constituency boundaries were redrawn using different values of the maximum post-treatment violation of parallel trends (Mbar). Each value on the horizontal axis is a different value of Mbar. The breakdown value corresponds to the first value of Mbar that results in a confidence set that has a statistically insignificant treatment effect.

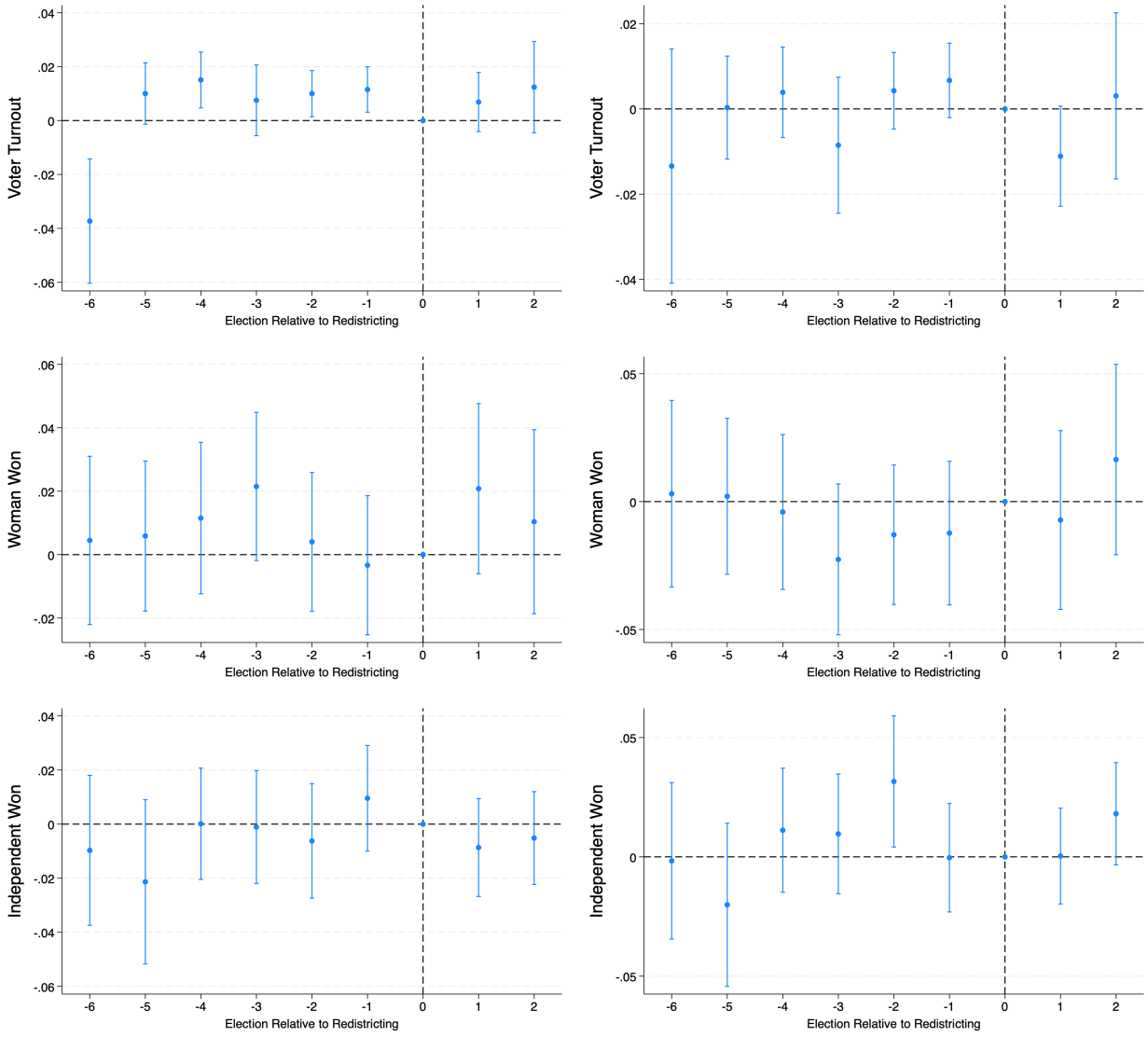
Figure A.8: Event Study Sensitivity Analysis: Effect of Reservation Status Change on Muslim Candidate Nominations



(a) New Voters (b) Reservation Status Change

Notes: This figure depicts the results of the event-study sensitivity analysis proposed in Rambachan and Roth (2023). Each panel plots the robust 95% confidence sets for the treatment effect in the election year after constituency boundaries were redrawn using different values of the maximum post-treatment violation of parallel trends (Mbar). Each value on the horizontal axis is a different value of Mbar. The breakdown value corresponds to the first value of Mbar that results in a confidence set that has a statistically insignificant treatment effect.

Figure A.9: Event Studies: Effect of Redistricting on Voter Behavior and Representation

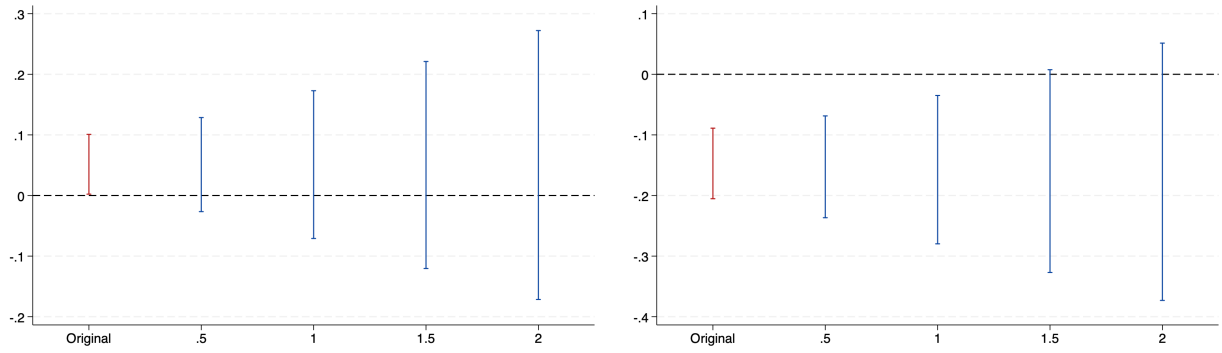


(a) Fraction of New Voters

(b) Reservation Status Change

Notes: Each point in Panel A represents the coefficient of $NewVoters_c$ interacted with an election year indicator variable from Equation 2. Each point in Panel B is the same but uses $ResChange_c$. The bars represent the 95% confidence interval of the coefficient. Each coefficient is relative to the year before redistricting. The vertical line is at the last election before electoral boundaries were redrawn.

Figure A.10: Event Study Sensitivity Analysis: Effect of Redistricting on the Probability an Incumbent Won



(a) New Voters

(b) Reservation Status Change

Notes: This figure depicts the results of the event-study sensitivity analysis proposed in Rambachan and Roth (2023). Each panel plots the robust 95% confidence sets for the treatment effect in the election year after constituency boundaries were redrawn using different values of the maximum post-treatment violation of parallel trends (Mbar). Each value on the horizontal axis is a different value of Mbar. The breakdown value corresponds to the first value of Mbar that results in a confidence set that has a statistically insignificant treatment effect.

Table A.1: Election Information by State

	First Election with New Maps	Election Years	Number of Elections
Andhra Pradesh	2009	1978-2014	9
Bihar	2010	1980-2015	8
Chhattisgarh	2008	2003-2013	3
Goa	2012	1989-2017	7
Gujurat	2012	1980-2017	9
Haryana	2009	1982-2014	8
Himachal Pradesh	2012	1982-2017	9
Karnataka	2008	1978-2013	9
Kerala	2011	1980-2016	9
Madhya Pradesh	2008	1980-2013	8
Maharashtra	2009	1978-2014	9
Meghalaya	2008	1978-2013	8
Mizoram	2008	1979-2013	9
Odisha	2009	1980-2014	8
Puducherry	2011	1980-2016	9
Punjab	2012	1980-2017	8
Rajasthan	2008	1980-2013	8
Sikkim	2009	1979-2014	8
Tamil Nadu	2011	1980-2016	9
Tripura	2008	1983-2013	7
Uttarakhand	2012	2002-2017	4
Uttar Pradesh	2012	1980-2017	10
West Bengal	2011	1982-2016	8

Notes: This table presents the election data for each state in India. Elections are scheduled every five years. Chhattisgarh split from Madhya Pradesh in 2000, and Uttarakhand split from Uttar Pradesh in 2000. I have manually followed the politicians in Chhattisgarh and Uttarakhand over time.

Table A.2: Pre-Redistricting Summary Statistics

	Observations (1)	Mean (2)	Std. dev. (3)	Minimum (4)	Maximum (5)
Panel A: Electoral Redistricting					
Fraction of New Voters in Constituency	3,235	0.329	0.224	0	0.999
Reservation Status Change	3,235	0.199	0.4	0	1
Panel B: Political Candidacy					
Incumbents Nominated by a Major Party	3,235	0.657	0.502	0	2
Women Nominated by a Major Party	3,235	0.329	0.584	0	6
Muslim Candidates Nominated by a Major Party	3,235	0.338	0.845	0	7
Independent Men	3,235	3.014	2.709	0	23
Independent Women	3,235	0.180	0.447	0	6
Panel C: Voter Behavior and Representation					
Voter Turnout	3,235	0.710	0.170	0.235	1
Any Incumbent Won	3,235	0.336	0.473	0	1
Woman Won	3,235	0.070	0.255	0	1
Minority Candidate Won	3,235	0.051	0.220	0	1
Independent Won	3,235	0.038	0.191	0	1
Panel D: Economic Development					
Primary Schools	2,436	87.251	14.690	0	100
Middle Schools	2,436	39.112	25.423	0	100
Primary Health Centers	2,436	7.702	13.404	0	100
Primary Health Subcenters	2,436	22.317	21.144	0	100
Paved Roads	2,436	68.635	26.840	0	100
Annual Nighttime Light Growth	2,840	7.484	16.936	-78.792	85.420

Notes: This table presents summary statistics of all of the main outcomes. Data is for the election immediately before redistricting

Table A.3: Type of Reservation Change - Political Candidacy

	Major Party Nominees				
	Number of Incumbents (1)	Number of Women (2)	Number of Muslim Candidates (3)	Number of Independent Men (4)	Number of Independent Women (5)
New Voters	0.095***	-0.018	-0.036	-0.206	-0.055
	(0.037)	(0.043)	(0.049)	(0.176)	(0.034)
Unreserved to Reserved	-0.376***	0.179**	-0.337***	-0.668***	0.058
	(0.050)	(0.071)	(0.060)	(0.237)	(0.052)
Reserved to Unreserved	-0.132*	-0.088	0.097	1.050**	0.061
	(0.073)	(0.085)	(0.103)	(0.416)	(0.059)
Observations	6,470	6,470	6,470	6,470	6,470
R-squared	0.561	0.579	0.721	0.684	0.534
Avg Dep Variable	0.657	0.329	0.338	3.014	0.180

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. "New Voters" is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. "Unreserved to Reserved" means a constituency was not a reserved seat for SC or ST candidates before redistricting and the seat became reserved due to redistricting. "Reserved to Unreserved" means a constituency was a reserved seat for SC or ST candidates before redistricting and the seat became unreserved due to redistricting.

Table A.4: Effect of Redistricting on Additional Candidacy Outcomes

	Number of Major Parties Contesting (1)	Number of Candidates (2)	Number of Incumbents (3)	Number of Women (4)	Number of Muslim Candidates (5)
$Post \times NewVoters$	-0.177	-0.332	0.086**	-0.056	-0.032
	(0.172)	(0.225)	(0.038)	(0.060)	(0.073)
$Post \times ResChange$	-0.051	-0.179	-0.279***	0.113	-0.316***
	(0.212)	(0.288)	(0.044)	(0.079)	(0.087)
Observations	6,470	6,470	6,470	6,470	6,470
R-squared	0.764	0.792	0.550	0.585	0.748
Avg Dep Variable	4.784	8.443	0.679	0.564	0.602

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.5: Robustness Check: Donut Specification - Political Candidacy

	Full Sample (1)	5 km (2)	10 km (3)	15 km (4)	20 km (5)
Panel A: Number of Major Party Incumbents					
$Post \times NewVoters$	0.096***	0.099***	0.092**	0.088**	0.043
	(0.037)	(0.038)	(0.039)	(0.042)	(0.046)
$Post \times ResChange$	-0.292***	-0.299***	-0.301***	-0.315***	-0.332***
	(0.043)	(0.044)	(0.044)	(0.048)	(0.054)
Panel B: Number of Major Party Women Candidates					
$Post \times NewVoters$	-0.019	-0.012	0.003	-0.010	-0.014
	(0.043)	(0.044)	(0.046)	(0.049)	(0.057)
$Post \times ResChange$	0.087	0.077	0.070	0.044	0.077
	(0.057)	(0.058)	(0.061)	(0.066)	(0.075)
Panel C: Number of Major Party Muslim Candidates					
$Post \times NewVoters$	-0.035	-0.041	-0.042	-0.051	-0.026
	(0.049)	(0.050)	(0.051)	(0.055)	(0.060)
$Post \times ResChange$	-0.187***	-0.192***	-0.204***	-0.242***	-0.243***
	(0.058)	(0.058)	(0.062)	(0.067)	(0.075)
Panel D: Number of Independent Men					
$Post \times NewVoters$	-0.201	-0.208	-0.186	-0.190	-0.153
	(0.177)	(0.181)	(0.190)	(0.206)	(0.236)
$Post \times ResChange$	-0.077	-0.085	-0.036	-0.091	0.045
	(0.222)	(0.226)	(0.239)	(0.263)	(0.302)
Panel E: Number of Independent Women					
$Post \times NewVoters$	-0.055	-0.054	-0.051	-0.056	-0.067
	(0.034)	(0.034)	(0.036)	(0.039)	(0.046)
$Post \times ResChange$	0.059	0.053	0.060	0.057	0.063
	(0.042)	(0.042)	(0.045)	(0.049)	(0.058)

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Column 1 is the full sample. Column 2 drops constituencies within 5 kilometers of the nearest treated constituency. Column 3 drops constituencies within 10 kilometers of the nearest treated constituency. Column 4 drops constituencies within 15 kilometers of the nearest treated constituency. Column 5 drops constituencies within 20 kilometers of the nearest treated constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.6: Candidacy Outcomes: Full Sample, Additional Controls, and 1-to-1 Mapping

	Major Party Nominees				
	Number of Incumbents (1)	Number of Women (2)	Number of Muslim Candidates (3)	Number of Independent Men (4)	Number of Independent Women (5)
Panel A: Full Sample					
$Post \times NewVoters$	0.083***	-0.014	-0.045	-0.315**	-0.038*
	(0.025)	(0.028)	(0.034)	(0.148)	(0.022)
$Post \times ResChange$	-0.280***	0.090**	-0.040	0.434**	0.030
	(0.028)	(0.037)	(0.039)	(0.184)	(0.028)
Panel B: Controlling for Population Change					
$Post \times NewVoters$	0.098***	-0.019	-0.035	-0.200	-0.055
	(0.037)	(0.043)	(0.049)	(0.177)	(0.034)
$Post \times ResChange$	-0.293***	0.087	-0.187***	-0.077	0.059
	(0.043)	(0.057)	(0.058)	(0.222)	(0.042)
Panel C: Controlling for Population Loss					
$Post \times NewVoters$	0.091**	-0.022	-0.023	-0.227	-0.048
	(0.039)	(0.044)	(0.048)	(0.187)	(0.035)
$Post \times ResChange$	-0.292***	0.087	-0.188***	-0.075	0.058
	(0.043)	(0.057)	(0.058)	(0.222)	(0.042)
Panel D: One to One Mapping					
$Post \times NewVoters$	0.071***	-0.004	-0.029	-0.169	-0.045*
	(0.027)	(0.031)	(0.036)	(0.130)	(0.025)
$Post \times ResChange$	-0.305***	0.074*	-0.200***	0.098	0.055*
	(0.033)	(0.044)	(0.046)	(0.177)	(0.032)
Avg Dep Variable	0.657	0.329	0.338	3.014	0.180

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. Panel A utilized 6 elections pre-redistricting and one election post-redistricting. The other panels use one election post-redistricting and one election pre-redistricting are utilized for the analysis. Panel B controls for the population change of a constituency. Panel C controls for the percent decline in a constituency's population. Panel D includes only a one to one mapping of pre- to post-redistricting constituencies. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.7: Robustness Check: Political Candidacy - Alternative Cutoffs

	Major Party Nominees				
	Number of Incumbents (1)	Number of Women (2)	Number of Muslim Candidates (3)	Number of Independent Men (4)	Number of Independent Women (5)
Continuous	0.055***	-0.019	-0.017	-0.123	-0.031*
	(0.019)	(0.021)	(0.026)	(0.088)	(0.016)
Fourth Quartile	0.149***	-0.042	-0.060	-0.206	-0.079*
	(0.052)	(0.059)	(0.075)	(0.253)	(0.045)
First Quartile	-0.016	-0.285***	-0.076	-0.175	-0.115*
	(0.088)	(0.087)	(0.107)	(0.344)	(0.065)
Drop Outliers	0.100***	-0.015	-0.032	-0.189	-0.053
	(0.037)	(0.043)	(0.050)	(0.178)	(0.034)
Demographic Change	0.096***	-0.108***	0.018	0.196	-0.026
	(0.035)	(0.041)	(0.049)	(0.168)	(0.033)
Avg Dep Variable	0.657	0.329	0.338	3.014	0.180

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Row 1 uses a continuous measure of the fraction of new voters. Row 2 compares the fourth quartile of the fraction of new voters to the rest of the sample. Row 3 compares the first quartile of the fraction of new voters to the rest of the sample. Row 4 drops the top 1% of constituencies based on the fraction of new voters. Row 5 uses demographic change as the measure of redistricting instead of the fraction of new voters. All regressions include $Post \times ResChange$.

Table A.8: Robustness Check: Political Candidacy - Clustering

	Major Party Nominees				
	Number of Incumbents (1)	Number of Women (2)	Number of Muslim Candidates (3)	Number of Independent Men (4)	Number of Independent Women (5)
$Post \times NewVoters$	0.096	-0.019	-0.035	-0.201	-0.055
	(0.026)***	(0.030)	(0.032)	(0.133)	(0.024)**
	{0.028}***	{0.031}	{0.032}	{0.138}	{0.024}**
	[0.029]***	[0.030]	[0.031]	[0.157]	[0.022]**
	|0.032|***	|0.034|	|0.037|	|0.163|	|0.021|***
$Post \times ResChange$	-0.292	0.087	-0.187	-0.077	0.059
	(0.031)***	(0.039)**	(0.044)***	(0.162)	(0.029)**
	{0.031}***	{0.041}**	{0.044}***	{0.166}	{0.029}**
	[0.033]***	[0.035]**	[0.046]***	[0.165]	[0.026]**
	|0.032|***	|0.031|***	|0.055|***	|0.132|	|0.021|***

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the electoral district level and reported in parenthesis. The second through fourth standard errors allow for arbitrary spatial correlation following Colella et al. (2023). The standard errors in curly brackets allow for spatial correlation for constituencies within 50 kilometers of each other. The standard errors in square brackets allow for spatial correlation for constituencies within 100 kilometers of each other. The standard errors in absolute value bars allow for spatial correlation for constituencies within 250 kilometers of each other. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.9: Effect of Redistricting on Additional Representation Outcomes

	Vote Margin (1)	BJP Won (2)	INC Won (3)
$Post \times NewVoters$	-0.000	-0.000	0.029
	(0.006)	(0.023)	(0.028)
$Post \times ResChange$	-0.003	0.005	0.028
	(0.008)	(0.027)	(0.035)
Observations	6,470	6,470	6,470
R-squared	0.622	0.699	0.616
Avg Dep Variable	0.111	0.221	0.270

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting. "BJP" stands for the Bharatiya Janata Party, and "INC" stands for Indian National Congress.

Table A.10: Type of Reservation Change - Voter Behavior and Representation

	Voter Turnout (1)	An Incumbent Won (2)	A Woman Won (3)	A Muslim Candidate Won (4)	An Independent Won (5)
New Voters	0.007	0.051	0.021	0.011	-0.009
	(0.007)	(0.032)	(0.018)	(0.013)	(0.012)
Unreserved to Reserved	-0.031***	-0.176***	0.029	-0.044***	-0.015
	(0.009)	(0.047)	(0.028)	(0.016)	(0.016)
Reserved to Unreserved	0.028***	-0.091	-0.077**	0.014	0.029
	(0.011)	(0.058)	(0.039)	(0.025)	(0.021)
Observations	6,470	6,470	6,470	6,470	6,470
R-squared	0.861	0.551	0.582	0.679	0.524
Avg Dep Variable	0.710	0.336	0.070	0.051	0.038

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. "New Voters" is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. "Unreserved to Reserved" means a constituency was not a reserved seat for SC or ST candidates before redistricting and the seat became reserved due to redistricting. "Reserved to Unreserved" means a constituency was a reserved seat for SC or ST candidates before redistricting and the seat became unreserved due to redistricting.

Table A.11: Robustness Check: Donut Specification - Voter Behavior and Representation

	Full Sample (1)	5 km (2)	10 km (3)	15 km (4)	20 km (5)
Panel A: Voter Turnout					
$Post \times NewVoters$	0.007	0.009	0.011	0.011	0.008
	(0.007)	(0.007)	(0.007)	(0.008)	(0.009)
$Post \times ResChange$	-0.011	-0.012	-0.011	-0.010	-0.007
	(0.007)	(0.007)	(0.008)	(0.009)	(0.010)
Panel B: Any Incumbent Won					
$Post \times NewVoters$	0.052	0.051	0.042	0.047	0.017
	(0.032)	(0.033)	(0.034)	(0.037)	(0.041)
$Post \times ResChange$	-0.147***	-0.145***	-0.144***	-0.135***	-0.181***
	(0.039)	(0.039)	(0.040)	(0.042)	(0.048)
Panel C: A Woman Won					
$Post \times NewVoters$	0.021	0.020	0.025	0.025	0.013
	(0.018)	(0.018)	(0.019)	(0.020)	(0.023)
$Post \times ResChange$	-0.007	-0.007	-0.013	-0.022	-0.007
	(0.023)	(0.023)	(0.024)	(0.025)	(0.028)
Panel D: A Muslim Candidate Won					
$Post \times NewVoters$	0.011	0.012	0.012	0.005	0.001
	(0.013)	(0.013)	(0.013)	(0.014)	(0.015)
$Post \times ResChange$	-0.024*	-0.023	-0.026*	-0.024	-0.018
	(0.014)	(0.015)	(0.015)	(0.016)	(0.016)
Panel E: An Independent Won					
$Post \times NewVoters$	-0.009	-0.008	-0.011	-0.009	-0.012
	(0.012)	(0.012)	(0.013)	(0.014)	(0.016)
$Post \times ResChange$	0.000	-0.002	-0.002	-0.002	0.001
	(0.014)	(0.014)	(0.014)	(0.016)	(0.019)

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Column 1 is the full sample. Column 2 drops constituencies within 5 kilometers of the nearest treated constituency. Column 3 drops constituencies within 10 kilometers of the nearest treated constituency. Column 4 drops constituencies within 15 kilometers of the nearest treated constituency. Column 5 drops constituencies within 20 kilometers of the nearest treated constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.12: Representation: Full Sample, Additional Controls, and 1-to-1 Mapping

	Voter Turnout (1)	An Incumbent Won (2)	A Woman Won (3)	A Muslim Candidate Won (4)	An Independent Won (5)
Panel A: Full Sample					
$Post \times NewVoters$	0.001	0.048**	0.014	0.013	-0.006
	(0.006)	(0.021)	(0.012)	(0.008)	(0.007)
$Post \times ResChange$	-0.011*	-0.132***	0.001	-0.011	-0.004
	(0.006)	(0.024)	(0.014)	(0.009)	(0.008)
Panel B: Controlling for Population Change					
$Post \times NewVoters$	0.007	0.054*	0.021	0.011	-0.009
	(0.007)	(0.033)	(0.018)	(0.013)	(0.012)
$Post \times ResChange$	-0.011	-0.148***	-0.007	-0.024*	0.000
	(0.007)	(0.039)	(0.023)	(0.014)	(0.014)
Panel C: Controlling for Population Loss					
$Post \times NewVoters$	0.007	0.049	0.015	0.014	-0.008
	(0.007)	(0.034)	(0.018)	(0.014)	(0.013)
$Post \times ResChange$	-0.011	-0.147***	-0.007	-0.024*	0.000
	(0.007)	(0.039)	(0.023)	(0.014)	(0.014)
Panel D: One to One Mapping					
$Post \times NewVoters$	0.007	0.049**	0.016	0.018*	-0.006
	(0.005)	(0.024)	(0.013)	(0.010)	(0.009)
$Post \times ResChange$	-0.012**	-0.169***	-0.006	-0.026**	-0.007
	(0.006)	(0.029)	(0.018)	(0.011)	(0.011)
Avg Dep Variable	0.710	0.336	0.070	0.051	0.038

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. Panel A utilized 6 elections pre-redistricting and one election post-redistricting. The other panels use one election post-redistricting and one election pre-redistricting are utilized for the analysis. Panel B controls for the population change of a constituency. Panel C controls for the percent decline in a constituency's population. Panel D includes only a one to one mapping of pre- to post-redistricting constituencies. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.13: Robustness Check: Voter Behavior and Representation - Alternative Cutoffs

	Voter Turnout (1)	An Incumbent Won (2)	A Woman Won (3)	A Muslim Candidate Won (4)	An Independent Won (5)
Continuous	0.003	0.024	0.009	0.001	-0.003
	(0.003)	(0.016)	(0.009)	(0.007)	(0.006)
Fourth Quartile	0.007	0.061	0.020	0.007	-0.010
	(0.009)	(0.045)	(0.025)	(0.019)	(0.017)
First Quartile	0.002	0.039	-0.046	-0.001	0.026
	(0.014)	(0.078)	(0.037)	(0.026)	(0.029)
Drop Outliers	0.007	0.052	0.021	0.013	-0.010
	(0.007)	(0.033)	(0.018)	(0.013)	(0.012)
Demographic Change	0.008	0.001	-0.015	0.017	0.007
	(0.006)	(0.031)	(0.017)	(0.013)	(0.012)
Avg Dep Variable	0.710	0.336	0.070	0.051	0.038

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Row 1 uses a continuous measure of the fraction of new voters. Row 2 compares the fourth quartile of the fraction of new voters to the rest of the sample. Row 3 compares the first quartile of the fraction of new voters to the rest of the sample. Row 4 drops the top 1% of constituencies based on the fraction of new voters. Row 5 uses demographic change as the measure of redistricting instead of the fraction of new voters. All regressions include $Post \times ResChange$.

Table A.14: Robustness Check: Voter Behavior and Representation - Clustering

	Voter Turnout (1)	An Incumbent Won (2)	A Woman Won (3)	A Muslim Candidate Won (4)	An Independent Won (5)
$Post \times NewVoters$	0.007	0.052	0.021	0.011	-0.009
	(0.005)	(0.022)**	(0.014)	(0.009)	(0.009)
	{0.005}	{0.024}**	{0.014}	{0.009}	{0.009}
	[0.005]	[0.023]**	[0.014]	[0.008]	[0.009]
	|0.005|	|0.027|*	|0.013|	|0.011|	|0.010|
$Post \times ResChange$	-0.011	-0.147	-0.007	-0.024	0.000
	(0.005)**	(0.029)***	(0.016)	(0.009)***	(0.009)
	{0.006}**	{0.028}***	{0.016}	{0.010}**	{0.009}
	[0.006]*	[0.029]***	[0.016]	[0.012]**	[0.009]
	|0.008|	|0.025|***	|0.015|	|0.012|**	|0.010|

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the electoral district level and reported in parenthesis. The second through fourth standard errors allow for arbitrary spatial correlation following Colella et al. (2023). The standard errors in curly brackets allow for spatial correlation for constituencies within 50 kilometers of each other. The standard errors in square brackets allow for spatial correlation for constituencies within 100 kilometers of each other. The standard errors in absolute value bars allow for spatial correlation for constituencies within 250 kilometers of each other. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.15: Type of Reservation Change - Development Outcomes

	Primary Schools (1)	Middle Schools (2)	Health Centers (3)	Health Subcenters (4)	Paved Roads (5)	Luminosity Growth (6)
New Voters	-1.249*	-1.268	-0.577	0.095	0.836	-0.016
	(0.682)	(0.939)	(0.412)	(0.867)	(1.030)	(0.344)
Unreserved to Reserved	-1.328	0.503	1.488***	-0.207	0.425	0.013***
	(0.997)	(1.247)	(0.519)	(1.126)	(1.436)	(0.005)
Reserved to Unreserved	2.903**	3.934**	-0.174	0.201	3.316*	0.011
	(1.239)	(1.570)	(0.726)	(1.326)	(1.844)	(0.007)
Observations	4,872	4,872	4,872	4,872	4,872	5,680
R-squared	0.706	0.820	0.875	0.809	0.784	0.020
Avg Dep Variable	87.251	39.112	7.702	22.317	68.635	7.484

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. "New Voters" is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. "Unreserved to Reserved" means a constituency was not a reserved seat for SC or ST candidates before redistricting and the seat became reserved due to redistricting. "Reserved to Unreserved" means a constituency was a reserved seat for SC or ST candidates before redistricting and the seat became unreserved due to redistricting. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. The R-squared value in Column 6 is a pseudo-R-squared value from the PPML estimation.

Table A.16: Robustness Check: Same Sample Across All Main Outcomes

	Number of Major Party Incumbents (1)	Any Incumbent Won (2)	Primary Schools (3)	Middle Schools (4)	Health Centers (5)	Luminosity Growth (6)
$Post \times NewVoters$	0.112***	0.063*	-1.226*	-1.249	-0.586	-0.016
	(0.043)	(0.037)	(0.682)	(0.939)	(0.413)	(0.580)
$Post \times ResChange$	-0.299***	-0.141***	0.160	1.709*	0.904**	1.666***
	(0.049)	(0.043)	(0.822)	(1.038)	(0.451)	(0.584)
Observations	4,872	4,872	4,872	4,872	4,872	4,872
R-squared	0.564	0.549	0.705	0.820	0.875	0.017
Avg Dep Variable	0.653	0.323	87.251	39.112	7.702	5.949

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Panel A conducts the analysis on the same sample as the public goods results. Panel B conducts the analysis on the same sample as the nighttime lights results. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. The R-squared value in Column 6 is a pseudo-R-squared value from the PPML estimation.

Table A.17: Robustness Check: Donut Specification - Development Outcomes

	Full Sample (1)	5 km (2)	10 km (3)	15 km (4)	20 km (5)
Panel A: Primary Schools					
$Post \times NewVoters$	-0.012*	-0.010	-0.006	-0.009	-0.006
	(0.007)	(0.007)	(0.007)	(0.007)	(0.008)
$Post \times ResChange$	0.002	0.002	-0.000	0.005	0.013
	(0.008)	(0.008)	(0.008)	(0.009)	(0.010)
Panel B: Middle Schools					
$Post \times NewVoters$	-1.249	-1.061	-0.673	-1.785*	-2.039*
	(0.939)	(0.944)	(0.949)	(0.989)	(1.115)
$Post \times ResChange$	1.709*	1.576	1.190	1.476	2.120*
	(1.038)	(1.043)	(1.058)	(1.130)	(1.265)
Panel C: Primary Health Centers					
$Post \times NewVoters$	-0.006	-0.006	-0.003	-0.007*	-0.005
	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
$Post \times ResChange$	0.009**	0.010**	0.012***	0.012***	0.013***
	(0.005)	(0.004)	(0.004)	(0.004)	(0.004)
Panel D: Primary Health Subcenters					
$Post \times NewVoters$	0.001	-0.002	-0.002	-0.011	-0.013
	(0.009)	(0.009)	(0.009)	(0.009)	(0.010)
$Post \times ResChange$	-0.001	0.001	0.003	0.004	0.007
	(0.009)	(0.009)	(0.009)	(0.010)	(0.011)
Panel E: Paved Roads					
$Post \times NewVoters$	0.009	0.008	0.010	0.004	-0.006
	(0.010)	(0.010)	(0.011)	(0.011)	(0.013)
$Post \times ResChange$	0.014	0.014	0.008	0.017	0.022
	(0.012)	(0.012)	(0.012)	(0.013)	(0.015)
Panel F: Nighttime Light Growth					
$Post \times NewVoters$	-0.000	0.000	0.000	-0.001	0.001
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
$Post \times ResChange$	0.012***	0.009**	0.006*	0.004	0.005
	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Column 1 is the full sample. Column 2 drops constituencies within 5 kilometers of the nearest treated constituency. Column 3 drops constituencies within 10 kilometers of the nearest treated constituency. Column 4 drops constituencies within 15 kilometers of the nearest treated constituency. Column 5 drops constituencies within 20 kilometers of the nearest treated constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting.

Table A.18: Development Outcomes: Full Sample, Additional Controls, and 1-to-1 Mapping

	Primary Schools (1)	Middle Schools (2)	Health Centers (3)	Health Subcenters (4)	Paved Roads (5)	Luminosity Growth (6)
Panel A: Full Sample						
$Post \times NewVoters$	-0.947	-1.387	-0.357	-0.278	0.602	-0.306
	(0.623)	(1.137)	(0.900)	(1.140)	(0.967)	(0.230)
$Post \times ResChange$	0.253	3.241***	2.079**	0.118	3.038***	0.632**
	(0.750)	(1.230)	(1.000)	(1.201)	(1.098)	(0.284)
Panel B: Controlling for Population Change						
$Post \times NewVoters$	-0.846	-0.975	-0.531	-0.329	0.836	0.002
	(0.714)	(0.963)	(0.410)	(0.859)	(1.040)	(0.344)
$Post \times ResChange$	0.120	1.681	0.898**	-0.019	1.443	1.215***
	(0.821)	(1.038)	(0.450)	(0.914)	(1.195)	(0.401)
Panel C: Controlling for Population Loss						
$Post \times NewVoters$	-1.584**	-1.686*	-0.976**	-0.138	0.882	-0.641*
	(0.761)	(1.012)	(0.483)	(0.914)	(1.086)	(0.338)
$Post \times ResChange$	0.194	1.751*	0.941**	-0.041	1.439	1.272***
	(0.825)	(1.041)	(0.454)	(0.914)	(1.196)	(0.406)
Panel D: One to One Mapping						
$Post \times NewVoters$	-1.104	-0.840	-0.518	0.190	1.329	0.014
	(0.715)	(0.963)	(0.416)	(0.859)	(1.080)	(0.418)
$Post \times ResChange$	0.309	1.576	1.040**	-0.205	2.115	1.116**
	(0.895)	(1.101)	(0.495)	(0.963)	(1.312)	(0.537)
Avg Dep Variable	87.251	39.112	7.702	22.317	68.635	7.484

Notes: $^*p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. Panel A utilized 6 elections pre-redistricting and one election post-redistricting. The other panels use one election post-redistricting and one election pre-redistricting are utilized for the analysis. Panel B controls for the population change of a constituency. Panel C controls for the percent decline in a constituency's population. Panel D includes only a one to one mapping of pre- to post-redistricting constituencies. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. The R-squared value in Column 6 is a pseudo-R-squared value from the PPML estimation.

Table A.19: Robustness Check: Development Outcomes - Alternative Cutoffs

	Primary Schools (1)	Middle Schools (2)	Health Centers (3)	Health Subcenters (4)	Paved Roads (5)	Luminosity Growth (6)
Continuous	-1.016***	-1.152**	-0.400*	0.126	0.138	-0.000
	(0.350)	(0.498)	(0.236)	(0.475)	(0.506)	(0.002)
Fourth Quartile	-2.222***	-2.289**	-0.633	-0.429	-0.102	-0.001
	(0.760)	(1.112)	(0.560)	(1.063)	(1.123)	(0.005)
First Quartile	-1.587	-2.495	-0.080	0.877	4.674*	-0.009
	(2.225)	(3.186)	(1.831)	(2.443)	(2.567)	(0.012)
Drop Outliers	-1.123*	-1.164	-0.533	-0.007	0.864	0.043
	(0.682)	(0.941)	(0.411)	(0.867)	(1.040)	(0.346)
Demographic Change	-1.286*	-1.803**	-0.079	0.738	1.256	-0.017***
	(0.668)	(0.912)	(0.406)	(0.843)	(0.999)	(0.004)
Avg Dep Variable	87.251	39.112	7.702	22.317	68.635	7.484

Notes: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. Row 1 uses a continuous measure of the fraction of new voters. Row 2 compares the fourth quartile of the fraction of new voters to the rest of the sample. Row 3 compares the first quartile of the fraction of new voters to the rest of the sample. Row 4 drops the top 1% of constituencies based on the fraction of new voters. Row 5 uses demographic change as the measure of redistricting instead of the fraction of new voters. All regressions include $Post \times ResChange$. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth.

Table A.20: Robustness Check: Development Outcomes - Clustering

	Primary Schools (1)	Middle Schools (2)	Health Centers (3)	Health Subcenters (4)	Paved Roads (5)	Luminosity Growth (6)
$Post \times NewVoters$	-1.226	-1.249	-0.586	0.097	0.852	-0.016
	(0.752)	(1.020)	(0.395)	(0.933)	(1.095)	(0.347)
	{0.762}	{1.089}	{0.458}	{0.994}	{1.139}	
	[0.785]	[1.237]	[0.263]**	[0.949]	[1.204]	
	|0.736|*	|1.227|	|0.268|**	|0.885|	|1.144|	
$Post \times ResChange$	0.160	1.709	0.904	-0.063	1.442	1.221
	(1.101)	(1.184)	(0.521)*	(1.010)	(1.216)	(0.473)***
	{1.152}	{1.268}	{0.515}*	{1.033}	{1.276}	
	[1.287]	[1.430]	[0.531]*	[1.027]	[1.278]	
	|1.628|	|1.663|	|0.560|	|0.862|	|1.525|	

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the electoral district level and reported in parenthesis. The second through fourth standard errors allow for arbitrary spatial correlation following Colella et al. (2023). The standard errors in curly brackets allow for spatial correlation for constituencies within 50 kilometers of each other. The standard errors in square brackets allow for spatial correlation for constituencies within 100 kilometers of each other. The standard errors in absolute value bars allow for spatial correlation for constituencies within 250 kilometers of each other. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. *NewVoters* is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. *ResChange* is an indicator equal to one if a constituency's reservation status changed due to redistricting. Column 6 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. Column 6 does not include standard errors for arbitrary spatial correlation as the arbitrary spatial correlation is for linear functions.

Table A.21: Robustness Check: Nighttime Light Growth - Low Sensitivity and Saturation

	Luminosity Growth (1)
$Post \times NewVoters$	-0.299
	(0.302)
$Post \times ResChange$	0.650*
	(0.347)
Observations	5,464
R-squared	0.013
Avg Dep Variable	7.751

Notes: *$p < 0.1$, **$p < 0.05$, ***$p < 0.01$. Robust standard errors are clustered at the post-redistricting electoral constituency level. Each regression includes pre-redistricting electoral constituency and time fixed effects. One election post-redistricting and one election pre-redistricting are utilized for the analysis. The unit of analysis is the electoral constituency. $NewVoters$ is an indicator equal to one if a constituency had the fraction of new voters in the constituency at or above the median across India. $ResChange$ is an indicator equal to one if a constituency's reservation status changed due to redistricting. Column 1 uses a Poisson pseudo maximum likelihood (PPML) estimator to estimate the effect of redistricting on nighttime light growth. The R-squared value in Column 1 is a pseudo-R-squared value from the PPML estimation. The top and bottom 1% of the nighttime light constituencies are dropped.